

Volatility Persistence as a Financial State Variable: Memory, Roughness, and Forecasting Across Market Regimes

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Abstract

This paper asks whether volatility persistence carries forecasting information as a shared financial state beyond what standard benchmarks already contain. Using 115 equities drawn from the S&P 500 over November 2001–April 2026, we estimate rolling long-memory (GPH and local-Whittle) and roughness measures on Parkinson range-based variance and evaluate them in a walk-forward, point-in-time design against HAR and HAR-X (HAR with VIX and MOVE) benchmarks, using panel-aware Diebold–Mariano and Clark–West tests. Volatility proxies display strong persistence and low Hurst exponents, and the cross-sectional persistence state nearly doubles in the 2008 and 2020 crises and co-moves with the VIX. Its forecasting content, however, is largely redundant with the market-state controls: the full persistence specification improves on HAR by 4.5–8.3% at daily and weekly horizons but never significantly on HAR-X; pre-specified variants that add persistence to HAR-X directly do not lower forecast error; pooled estimation recovers only a negligible gain; and the persistence state predicts neither the term-structure slope of future variance nor market-level variance beyond VIX and MOVE. Volatility-managed portfolio differences are not statistically significant. Persistence is a coherent descriptive market state whose forecasting information is already contained in implied volatility.

Keywords: long memory; rough volatility; volatility forecasting; financial econometrics; realized volatility; persistence dynamics

JEL Classification: C22; C53; G12; G17

1. Introduction

Volatility forecasting occupies a central role in financial econometrics because volatility enters risk management, derivative pricing, portfolio allocation, margin determination, stress testing, and macro-financial monitoring. A large empirical literature has documented that volatility measures exhibit substantially greater persistence than raw asset returns. Classical long-memory models formalized this observation by extending conventional weak-dependence frameworks to allow autocorrelations that decay at hyperbolic rather than exponential rates (Granger & Joyeux, 1980; Hosking, 1981). Subsequent work incorporated fractional dependence into conditional-volatility specifications through FIGARCH-type models (Baillie et al., 1996), while the realized-volatility literature improved the empirical measurement of latent volatility dynamics using high-frequency data (Andersen et al., 2003; Barndorff-Nielsen & Shephard, 2002).

Despite extensive empirical evidence for persistence in volatility proxies, the interpretation of persistence remains controversial. One branch of the literature views long-

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memory behavior as reflecting structural characteristics of financial markets, including heterogeneous trading horizons, slow information diffusion, and institutional frictions (Müller et al., 1997). Another branch argues that apparent long-range dependence may arise from structural breaks, stochastic regime changes, or nonstationary volatility dynamics rather than genuine fractional dependence (Mikosch & Stărică, 2004). As a result, the economic interpretation of persistence measures remains unsettled. The central issue is therefore not simply whether a fractional parameter can be estimated, but whether persistence measures behave coherently across market states and contain useful information for forecasting and risk management.

This paper examines volatility persistence from a reduced-form perspective. Rather than interpreting the memory parameter as a purely technical object associated with fractional differencing, we investigate whether rolling persistence measures behave empirically like indicators of the temporal propagation of uncertainty in financial markets. More specifically, we study whether persistence measures rise during sustained stress episodes, whether they co-move with broad market uncertainty indicators, and whether they improve volatility forecasts beyond standard benchmark models. The analysis does not attempt to identify a unique structural mechanism generating persistence. Instead, persistence is treated as an empirical summary statistic whose dynamics may reflect several overlapping mechanisms, including heterogeneous investor horizons, liquidity conditions, delayed information processing, regime shifts, and market-wide stress propagation.

The paper also considers the relationship between long-memory behavior and rough volatility. Recent research has shown that volatility paths often display substantial local irregularity, with estimated Hurst exponents significantly below the Brownian benchmark of $\frac{1}{2}$ (Bennedsen et al., 2022; Gatheral et al., 2018). Although roughness and long memory are sometimes presented as competing explanations of volatility dynamics, they describe different temporal characteristics. Long memory concerns the persistence of dependence over long horizons, whereas roughness concerns local path irregularity at very short horizons. The empirical analysis in this paper suggests that both characteristics may coexist within the same volatility process.

A further issue concerns the role of machine learning in volatility forecasting. Flexible forecasting algorithms can capture nonlinear interactions and high-dimensional predictor structures that may be difficult to represent using conventional parametric models. However, purely data-driven forecasting systems often lack economic interpretability and may produce unstable results outside the estimation sample (Gu et al., 2020). The present paper adopts an intermediate approach: econometric methods are used to construct interpretable persistence-related features, while forecasting systems incorporate these features alongside conventional volatility predictors and market-state variables.

The empirical analysis uses a panel of 115 S&P 500 stocks over the period November 2001 through April 2026. We estimate rolling long-memory measures using both the Geweke–Porter–Hudak and local-Whittle estimators and estimate local roughness using rolling Hurst exponents. These variables are incorporated into a forecasting framework that compares heterogeneous autoregressive (HAR) models, HAR models with market-state controls (HAR-X), persistence-augmented regressions, shrinkage estimators, and tree-based machine-learning methods.

The central finding is negative in a specific and, we argue, useful way. Own-stock persistence and roughness add nothing to HAR. The shared persistence state, meaning cross-sectional and sectoral averages of rolling memory estimates and their interactions with market stress, does improve on HAR, but its improvement is almost entirely the improvement that VIX and MOVE deliver on their own. Tested directly against HAR-X with the appropriate nested-model statistics, no persistence specification lowers forecast

error at any horizon, and a set of follow-up designs fixed in advance (adding persistence blocks to HAR-X one at a time, letting persistence modulate the HAR weights, pooled estimation across stocks, a term-structure target, and a market-level target) shows that the information the persistence state carries is small and already contained in implied volatility.

The paper makes three contributions. First, it documents persistent low-frequency dependence alongside roughness diagnostics across a broad equity panel, distinguishing these temporal characteristics without treating either as proof of a particular stochastic law, and shows that a cross-sectional persistence state is a coherent market variable that nearly doubles in the global financial crisis and in COVID and co-moves with the VIX. Second, it provides a complete, point-in-time forecasting evaluation of persistence at the stock, cross-sectional, sectoral, and interaction levels against both HAR and HAR-X, with panel-aware Diebold–Mariano and Clark–West inference and pre-specified follow-up designs, and reports the direct comparisons with HAR-X that work on persistence features typically omits. Third, it relates this evidence to a volatility-managed portfolio application with formal tests of Sharpe-ratio differences and compares linear, shrinkage, and tree-based estimators on the same feature space. The reported results favor structured linear forecasts in this setting, not a general rejection of nonlinear methods.

The scope of the evidence is deliberately reduced-form. Persistence rises during major stress episodes and co-moves with the VIX, but the lead–lag diagnostic does not establish a leading stress signal, and the persistence state does not forecast how slowly variance decays. Every specification is tested against both HAR and HAR-X. The contribution is therefore a characterization: shared persistence states are informative descriptions of market conditions whose forecasting content, for stock-level variance, is captured by implied-volatility indices.

The remainder of the paper proceeds as follows. Section 2 reviews the literature. Section 3 discusses the theoretical foundations of persistence. Section 4 presents the econometric framework. Section 5 develops the reduced-form interpretation. Section 6 introduces the forecasting framework. Section 7 describes the data and empirical design. Section 8 presents the results. Section 9 discusses the economic implications. Section 10 concludes.

2. Literature Review: Persistence, Roughness, and Volatility

The literature on volatility persistence originates with foundational work on fractional integration and ARFIMA processes, which extended classical time-series analysis by allowing autocorrelation structures with hyperbolically decaying dependence rather than conventional exponential decay (Granger & Joyeux, 1980; Hosking, 1981). These developments became especially important in financial econometrics because volatility proxies such as squared returns, absolute returns, and realized variance measures exhibit substantially stronger persistence than raw asset returns themselves (Ding et al., 1993). The resulting evidence motivated the development of models capable of capturing long-range dependence in financial volatility while remaining consistent with broader conditional heteroskedasticity frameworks.

One important contribution in this direction was the FIGARCH model introduced by Baillie et al. (1996), which incorporated fractional integration directly into conditional variance dynamics. FIGARCH generalized conventional GARCH specifications by allowing shocks to decay at hyperbolic rates, thereby accommodating the empirically observed persistence of volatility. The model rapidly became a standard benchmark in long-memory volatility research because it provided a tractable parametric representation of persistent conditional heteroskedasticity. Subsequent extensions introduced asymmetric responses,

multivariate formulations, and alternative fractional structures, further expanding the long-memory volatility literature.

At approximately the same time, the realized-volatility literature substantially improved the empirical measurement of latent volatility processes. Andersen et al. (2003) and Barndorff-Nielsen and Shephard (2002) demonstrated that high-frequency financial data could be used to construct more accurate ex post measures of integrated volatility. These developments strengthened empirical evidence for volatility persistence because realized-volatility measures contain less measurement noise than traditional squared-return proxies. The resulting data environment allowed researchers to study persistence properties with considerably greater precision.

Another important development was the HAR model proposed by Corsi (2009). Rather than imposing exact fractional integration, the HAR framework approximates long-memory behavior using a parsimonious multi-horizon structure involving daily, weekly, and monthly realized-volatility components. The HAR model proved empirically successful because it combines forecasting accuracy with economic interpretability. In particular, it is consistent with the idea that financial markets contain heterogeneous agents operating at different investment horizons (Müller et al., 1997). The HAR framework remains one of the most widely used benchmarks in volatility forecasting.

Despite substantial empirical support for persistence, the interpretation of persistence remains controversial. One branch of the literature views long memory as reflecting genuine structural characteristics of financial markets, including heterogeneous trading frequencies, delayed information diffusion, and institutional frictions. Another branch argues that apparent long-range dependence may arise from structural breaks, regime changes, stochastic volatility shifts, or nonlinear dynamics rather than from genuine fractional integration (Mikosch & Stărică, 2004). This critique implies that estimated persistence parameters should be interpreted cautiously because they may reflect evolving market conditions rather than stable stochastic mechanisms.

The debate surrounding long memory became even more important following the emergence of the rough-volatility literature. Gatheral et al. (2018) showed that volatility paths frequently exhibit substantial local irregularity, with estimated Hurst exponents significantly below the Brownian-motion benchmark of $H = \frac{1}{2}$. Subsequent work clarified that roughness and long memory are not necessarily competing concepts. Bennedsen et al. (2022) argued that short-run roughness and long-horizon persistence may represent distinct temporal features of volatility processes. The empirical analysis adopts this perspective by estimating both rolling persistence measures and rolling Hurst exponents within the same forecasting framework.

In parallel with these econometric developments, machine learning has become increasingly important in volatility forecasting and empirical asset pricing. Flexible learning algorithms are capable of capturing nonlinear interactions, threshold effects, and high-dimensional predictor structures (Breiman, 2001; Friedman, 2001; Gu et al., 2020; Hastie et al., 2009). However, purely data-driven approaches may produce unstable results outside the estimation sample and frequently lack economic interpretability.

This tension between statistical flexibility and economic interpretation has become a central issue in modern empirical finance. Recent research increasingly emphasizes hybrid approaches in which economically motivated features are combined with flexible forecasting architectures. In this setting, the quality and interpretability of the feature space may be more important than the complexity of the forecasting algorithm itself.

Taken together, this literature motivates a focused empirical question: does estimating persistence add forecasting information beyond the multi-horizon dynamics captured by HAR and the market conditions captured by HAR-X? The present paper addresses this

question by separating own-stock features from cross-sectional aggregates, sectoral states, and persistence–stress interactions. Its contribution is this comparative assessment of persistence as a forecasting state variable, rather than a new fractional model or identification of an information-diffusion mechanism.

3. Theoretical Foundations of Long Memory and Information Persistence

The concept of long memory emerged from attempts to describe stochastic processes whose dependence structures cannot be adequately represented by conventional short-memory models. Long-memory processes retain measurable dependence over substantially longer horizons, typically characterized through a slowly decaying autocovariance structure in which dependence declines at a hyperbolic rather than exponential rate (Granger & Joyeux, 1980; Hosking, 1981).

Formally, a covariance-stationary long-memory process may be characterized by an autocovariance function satisfying

$$\gamma(k) \sim c_\gamma k^{2d-1} \quad \text{as } k \rightarrow \infty, \quad (1)$$

where $d \in (0, \frac{1}{2})$ denotes the memory parameter.

Because $2d - 1 > -1$, the autocovariances are not absolutely summable, implying persistent dependence across long horizons. Equivalent characterizations arise in the frequency domain,

$$f(\lambda) \sim c_f \lambda^{-2d}, \quad \lambda \rightarrow 0, \quad (2)$$

providing the foundation for semiparametric estimators of long memory, including the Geweke–Porter–Hudak and local-Whittle procedures employed later in the paper (Robinson, 1995).

The fractional-difference operator $(1 - L)^d$ provides the bridge between these probabilistic properties and empirically estimable time-series models. Fractionally integrated ARFIMA systems express current observations as weighted combinations of past innovations, where the weights decay gradually rather than disappearing rapidly. Relative to standard ARMA structures, fractional integration allows shocks to persist over much longer horizons while preserving covariance stationarity when $d < \frac{1}{2}$. In financial econometrics, this framework proved especially useful because volatility measures routinely display persistence patterns that are inconsistent with short-memory dynamics.

Although long-memory models originated as statistical descriptions of temporal dependence, their interpretation in financial applications naturally raises broader economic questions. When persistence is observed in volatility proxies rather than in returns themselves, the memory parameter may be viewed as summarizing the duration over which uncertainty shocks continue to influence market behavior. Importantly, this interpretation should not be understood as structural identification of a unique mechanism. Rather, persistence measures are best viewed as reduced-form summaries reflecting several overlapping economic forces that jointly influence the temporal propagation of volatility.

Several economic mechanisms could generate persistent volatility. Heterogeneous trading horizons can aggregate into slow adjustment (Corsi, 2009; Müller et al., 1997); delayed information processing, liquidity constraints, and gradual portfolio rebalancing may also prolong responses to shocks. These channels motivate studying persistence, but the empirical design does not distinguish among them. Regime changes and structural breaks remain alternative explanations for similar low-frequency behavior (Mikosch & Stărică, 2004).

These considerations suggest that persistence measures may serve as reduced-form indicators of the temporal organization of uncertainty in financial markets. A stylized representation may be written as

$$I_t = \sum_{k=0}^{\infty} w_k \varepsilon_{t-k}, \quad w_k \sim c k^{d-1}, \quad (3)$$

where I_t denotes an uncertainty-related process influenced by past informational shocks ε_{t-k} , and d governs the speed at which the influence of past shocks declines over time.

It is important, however, to distinguish carefully between statistical persistence and economic interpretation. The presence of estimated long memory does not necessarily imply that markets literally follow stationary fractional stochastic laws. Structural breaks, regime changes, stochastic volatility shifts, and evolving institutional environments may generate empirical behavior resembling long-range dependence. Consequently, persistence measures should not be interpreted mechanically as identifying a single structural mechanism. Instead, they should be viewed as empirical indicators whose behavior may reflect several interacting economic channels simultaneously.

The interpretation developed in this paper therefore remains intentionally reduced-form. The central empirical question is not whether a unique structural mechanism can be identified, but rather whether persistence measures behave coherently across market states and whether they contain useful forecasting information beyond conventional volatility predictors. If persistence measures rise systematically during sustained stress episodes, co-move with broad uncertainty indicators, and improve volatility forecasts during crisis periods, then they may provide economically informative summaries of market conditions even without full structural identification.

This perspective also accommodates the rough-volatility literature. Roughness and long memory describe different aspects of temporal dependence rather than mutually exclusive mechanisms. Long memory concerns low-frequency dependence and the persistence of shocks across longer horizons. Roughness concerns local path irregularity at very short horizons. A rough-volatility process satisfies

$$E|\log \sigma_{t+\Delta}^2 - \log \sigma_t^2|^q \propto \Delta^{qH}, \quad (4)$$

where $H < \frac{1}{2}$ implies rougher behavior than standard Brownian motion (Gatheral et al., 2018). A volatility process may therefore be simultaneously rough at short horizons and persistent at longer horizons. Financial markets may react sharply and irregularly immediately after news arrivals while still exhibiting prolonged uncertainty propagation over subsequent periods.

The coexistence of roughness and persistence suggests that volatility dynamics should be analyzed across multiple temporal scales rather than through a single dependence measure. High-frequency volatility fluctuations may primarily reflect microstructure effects and rapid information arrivals, whereas medium- and long-horizon persistence may reflect broader institutional adjustment processes, funding conditions, and macro-financial uncertainty propagation. The empirical framework developed later in the paper incorporates both dimensions simultaneously through rolling estimates of long-memory parameters and rolling Hurst exponents.

For forecasting, the relevant question is whether estimated persistence summarizes dependence not already captured by volatility lags and market-state controls. The empirical sections investigate this through forecast comparisons across horizons and regimes. These comparisons assess predictive usefulness, not the expected duration of an identified shock.

4. Econometric Modeling of Fractional and Rough Volatility

The econometric analysis of volatility persistence has evolved along two closely related but conceptually distinct directions. The first focuses on long-memory dynamics and emphasizes slow hyperbolic decay over medium and long horizons. The second focuses on rough volatility and emphasizes highly irregular local behavior at very short horizons. The long-memory literature developed primarily within discrete-time econometrics through fractional integration and ARFIMA-type models. Rough-volatility models, by contrast, emerged from continuous-time stochastic-volatility theory and high-frequency financial econometrics. Recent empirical evidence suggests that these two perspectives are not mutually exclusive. The empirical framework developed in this paper therefore incorporates both dimensions in a common forecasting framework.

4.1. ARFIMA, GARCH, and FIGARCH

The ARFIMA framework provides a discrete-time representation of fractional dependence. For a covariance-stationary process $\{X_t\}$,

$$\phi(L)(1-L)^d X_t = \theta(L)\eta_t, \quad (5)$$

where $\phi(L)$ and $\theta(L)$ are finite-order lag polynomials, d denotes the fractional integration parameter, and η_t is a white-noise innovation process (Granger & Joyeux, 1980; Hosking, 1981). The operator $(1-L)^d$ generalizes conventional integer differencing to fractional orders and allows shocks to decay gradually over long horizons. When applied to volatility proxies such as squared returns, absolute returns, range-based volatility measures, or realized variance, ARFIMA models provide parsimonious representations of long-range dependence in financial volatility. Unlike standard short-memory models, fractional integration allows dependence to persist over extended periods while preserving covariance stationarity when $d < \frac{1}{2}$.

The conditional heteroskedasticity literature approached volatility persistence from a different perspective. Beginning with the ARCH model of Engle (1982) and the GARCH extension of Bollerslev (1986), volatility dynamics were modeled directly through evolving conditional variance equations. Standard GARCH systems successfully capture volatility clustering but imply geometric decay in the effects of shocks. Empirical volatility series, however, often display substantially slower decay than conventional GARCH structures permit. The FIGARCH model of Baillie et al. (1996) addressed this limitation by embedding fractional integration directly within the conditional-variance recursion,

$$[1 - \beta(L)] h_t = \omega + [1 - \beta(L) - \phi(L)(1-L)^d] \varepsilon_t^2, \quad (6)$$

where h_t denotes conditional variance and ε_t denotes return innovations. FIGARCH allows volatility shocks to decay hyperbolically rather than exponentially while preserving the conditional-heteroskedasticity structure.

The FIGARCH framework became particularly influential because it provided an explicit connection between long-memory econometrics and volatility modeling. Numerous subsequent extensions introduced asymmetric effects, multivariate specifications, nonlinear dynamics, and regime dependence. Nevertheless, the central empirical motivation remained unchanged: volatility shocks appear to dissipate much more slowly than conventional short-memory volatility models predict.

An important practical issue concerns interpretation. Although ARFIMA and FIGARCH systems estimate similar persistence behavior, they represent different modeling philosophies. ARFIMA-type systems emphasize the dependence structure of the observed process itself, whereas GARCH-type systems emphasize the evolution of latent

conditional variance. In practice, both approaches frequently generate evidence consistent with persistent volatility dynamics.

4.2. Realized Volatility and HAR Models

The development of realized-volatility methodology substantially improved empirical analysis of volatility persistence. Traditional volatility proxies based on squared daily returns contain substantial measurement noise and therefore obscure underlying dependence structures. High-frequency financial data made it possible to construct more accurate ex post measures of integrated volatility through realized-variance estimators (Andersen et al., 2003; Barndorff-Nielsen & Shephard, 2002). Realized-volatility measures significantly strengthened empirical evidence for persistent volatility dynamics because they provide less noisy approximations to latent volatility processes. These measures also facilitated more precise estimation of long-memory behavior and improved volatility forecasting performance across a wide range of applications.

An especially important contribution was the heterogeneous autoregressive (HAR) model proposed by Corsi (2009). The HAR framework approximates long-memory behavior using a parsimonious multi-horizon structure rather than imposing exact fractional integration,

$$\log RV_{t+1} = \beta_0 + \beta_d \log RV_t^{(d)} + \beta_w \log RV_t^{(w)} + \beta_m \log RV_t^{(m)} + u_{t+1}, \quad (7)$$

where $RV_t^{(d)}$, $RV_t^{(w)}$, and $RV_t^{(m)}$ denote daily, weekly, and monthly realized-volatility components. The HAR model proved remarkably successful empirically because it captures many observed features of long-memory volatility dynamics while remaining computationally simple and economically interpretable. The framework is closely connected to heterogeneous-agent theories in which market participants operate at different investment horizons (Müller et al., 1997). Daily components capture short-horizon trading activity, whereas weekly and monthly components reflect slower-moving institutional and macroeconomic adjustments.

The HAR framework also became a dominant benchmark in volatility forecasting because of its strong empirical performance relative to more complex nonlinear specifications. In the present paper, HAR models serve as the baseline forecasting architecture against which persistence-augmented models are evaluated.

4.3. Rough Volatility

The rough-volatility literature introduced a complementary perspective on volatility dynamics by emphasizing local path irregularity rather than long-horizon persistence. Gatheral et al. (2018) showed that volatility paths estimated from high-frequency financial data often display substantially rougher behavior than implied by standard Brownian-motion models. In particular, estimated Hurst exponents frequently lie well below the classical benchmark $H = \frac{1}{2}$.

A simplified rough-volatility representation may be written as

$$\log \sigma_t = \mu + \nu B_t^H, \quad (8)$$

where B_t^H denotes fractional Brownian motion with Hurst exponent $H < \frac{1}{2}$. From an econometric standpoint, roughness is commonly characterized through scaling relations involving moments of volatility increments,

$$E|\log \sigma_{t+\Delta}^2 - \log \sigma_t^2|^q \propto \Delta^{qH}, \quad (9)$$

where H governs the scaling of local volatility fluctuations across short horizons. Let $m(q, \Delta)$ denote the empirical q th absolute moment of log-volatility increments at lag Δ . We estimate H_t as the slope from regressing $\log m(q, \Delta)$ on $\log \Delta$ over a small grid of lags, divided by q .

The rough-volatility framework proved highly successful in explaining several empirical features of financial markets, including short-horizon volatility behavior and implied-volatility surface dynamics (Bennedsen et al., 2022). Rough models also provided improved consistency with observed option-market data and generated substantial interest in both academic and practitioner communities.

Importantly, roughness and long memory characterize different temporal properties of volatility. Roughness concerns highly local short-term irregularity, whereas long memory concerns persistence across broader temporal horizons. A volatility process may therefore be simultaneously rough and persistent. The empirical framework adopted in this paper explicitly incorporates both dimensions through rolling estimates of long-memory parameters and rolling Hurst exponents.

4.4. Semiparametric Estimation of Long Memory

Empirical estimation of long memory presents substantial econometric challenges because dependence structures must be distinguished from nonstationarity, regime shifts, and structural breaks. The literature therefore developed several semiparametric estimation procedures that focus specifically on low-frequency behavior.

One widely used approach is the Geweke–Porter–Hudak (GPH) log-periodogram estimator (Geweke & Porter-Hudak, 1983). The estimator exploits the low-frequency behavior of the spectral density by regressing the logarithm of the periodogram on transformed Fourier frequencies. Using the first m low-frequency ordinates, the estimator takes the form

$$\log I(\lambda_j) = \alpha - d \log[4 \sin^2(\lambda_j/2)] + u_j, \quad (10)$$

where $I(\lambda_j)$ is the periodogram at Fourier frequency $\lambda_j = 2\pi j/T$, $j = 1, \dots, m$, and T is the number of observations. The bandwidth is $m = \lfloor T^{0.65} \rfloor$. If $\hat{\beta}$ is the estimated slope on $\log[4 \sin^2(\lambda_j/2)]$, the displayed regression implies $\hat{d} = -\hat{\beta}$.

Another important procedure is the local-Whittle estimator proposed by Robinson (1995). Rather than relying on regression, the local-Whittle approach minimizes an approximation to the Gaussian likelihood over the low-frequency band of the spectrum. The estimator generally exhibits lower asymptotic variance under broad conditions and has become one of the standard semiparametric procedures in long-memory econometrics. Both estimators focus specifically on low-frequency dependence without imposing complete parametric ARMA or GARCH structures, making them attractive for empirical applications in which the short-memory component may be misspecified or unstable.

In addition to semiparametric methods, the present paper also uses FIGARCH estimates on representative subsets of stocks as parametric cross-checks. Combining semiparametric and parametric approaches helps reduce dependence on any single modeling framework and improves robustness of the empirical conclusions.

4.5. Identification Challenges

Several identification problems complicate empirical interpretation of persistence estimates. The first concerns structural nonstationarity. Repeated regime shifts, evolving volatility states, and structural breaks can generate empirical behavior resembling fractional dependence even when the underlying process is not truly fractionally integrated (Mikosch & Stărică, 2004). As a result, estimated persistence parameters should not automatically be interpreted as evidence for stationary long-memory laws.

The second challenge concerns the distinction between highly persistent GARCH systems and genuine fractional integration. Near-integrated GARCH models may produce empirical dynamics that are difficult to distinguish from FIGARCH processes in finite samples, especially when persistence parameters approach unity. A third challenge concerns rough-volatility estimation. High-frequency financial data contain substantial market microstructure noise arising from bid–ask bounce, asynchronous trading, discreteness effects, and liquidity frictions. Unless realized-volatility measures are carefully constructed, estimated roughness parameters may partly reflect measurement distortions rather than genuine volatility dynamics.

The present paper addresses these issues through several robustness procedures. Persistence is estimated on the Parkinson range-based variance proxy with both the GPH and local-Whittle estimators, FIGARCH estimates provide a parametric cross-check, rolling estimation windows of varying lengths are compared, and the forecasting target is alternatively defined from squared returns. The objective is not to establish exact structural identification of fractional stochastic laws, but rather to determine whether persistence-related measures behave coherently across market states and contain useful forecasting information.

These identification issues are central to the broader interpretation developed in the next section, where persistence measures are viewed as reduced-form indicators of uncertainty propagation rather than as mechanically interpreted fractional coefficients.

5. Persistence as a Reduced-Form Financial State Variable

The memory parameter d describes the decay of temporal dependence under a fractional model. In the empirical application, its rolling estimate $\hat{d}_{i,t}$ is used as a summary of recent dependence in a volatility proxy. A financial state variable here means an observable or estimated predictor summarizing market conditions; it does not imply a structurally identified or sufficient state representation. Estimated persistence can reflect fractional dependence, regime shifts, measurement effects, or several mechanisms operating together.

The central hypothesis concerns the level of aggregation at which persistence is informative. An own-stock estimate may largely duplicate the information in that stock's daily, weekly, and monthly volatility components. Cross-sectional and sectoral averages may summarize shared dependence less visible in those individual histories. Interactions with VIX and MOVE allow the association between persistence and future volatility to vary with market conditions. The forecasting design assesses these possibilities without assuming that each feature block must improve accuracy.

Three empirical patterns would be consistent with this interpretation. First, rolling persistence states may vary across calm and stressed periods. Second, they may co-move with market-wide uncertainty indicators while still contributing information conditional on those indicators. Third, additional forecasting value may depend on the horizon and regime. None of these patterns alone establishes a causal channel, and co-movement with stress is not equivalent to forecasting its onset or length.

The duration and propagation of uncertainty provide an economic interpretation rather than a directly estimated outcome. Heterogeneous adjustment, liquidity constraints, and delayed information processing can motivate slow dependence decay, as discussed in Section 3. However, this paper does not estimate shock half-lives, crisis exit probabilities, or the expected time until volatility normalizes. Consequently, $\hat{d}_{i,t}$ is not a calibrated measure of stress duration. Section 8.3 tests one implication of the duration interpretation directly: if persistence measured how long shocks linger, it should help forecast the slope of the future variance term structure. It does not, so the evidence is con-

sistent with uncertainty propagation only in the weak sense that persistence rises when stress rises.

The distinction from market-state controls is empirical, not absolute. VIX and MOVE already contain forward-looking volatility information, and HAR incorporates persistent multi-horizon dynamics. Persistence features must therefore be assessed against HAR-X rather than credited with the entire gain of a combined model over HAR. A full-model comparison with HAR-X measures the joint incremental contribution of the added feature set; it does not uniquely attribute that gain to cross-sectional averages, sectoral states, roughness, or individual interactions.

Roughness remains a complementary diagnostic of short-scale irregularity. Its inclusion alongside memory estimates allows both temporal dimensions to enter the forecasting exercise, but low estimated Hurst exponents do not by themselves establish incremental forecast value. The practical contribution is to evaluate persistence states conditional on established predictors, not to replace those predictors with a single parameter.

6. Structured Forecasting with Persistence-Based Features

If persistence measures contain economically meaningful information, the next question concerns how such information should be incorporated into volatility forecasting systems. The approach adopted in this paper treats persistence measures not as isolated econometric coefficients but as dynamic state variables describing the temporal structure of financial uncertainty. Forecasting systems are therefore constructed around economically interpretable persistence-based features rather than around purely mechanical lag structures.

The framework developed here occupies an intermediate position between classical econometric modeling and purely data-driven machine learning. Traditional parametric forecasting systems provide interpretability and structural discipline but may fail to capture nonlinear interactions and regime dependence. Purely flexible machine-learning systems may capture complex patterns while sacrificing economic transparency and out-of-sample stability (Gu et al., 2020). The objective of the present framework is therefore not to replace econometric structure with machine learning, but rather to combine economically interpretable feature construction with flexible forecasting architectures.

The central hypothesis is that persistence features may contain information not fully captured by HAR dynamics and VIX/MOVE controls. Whether that information is useful, and at which horizon, is an empirical question. The feature design allows shared persistence and its interaction with market stress to complement conventional predictors without assuming that persistence directly measures the remaining duration of uncertainty.

The forecasting framework organizes predictors into three groups: HAR time-aggregation components, market-state indicators (VIX and MOVE), and persistence-related features and interactions. These groups can contain overlapping information; their inclusion is not a structural decomposition of volatility into separately identified economic channels.

Formally, let $y_{t,h}$ denote the future volatility target at forecast horizon h , let $\mathbf{X}_t$ denote a vector of conventional predictors, and let $\mathbf{Z}_t$ denote a vector of persistence-based features. The forecasting problem may be written as

$$y_{t,h} = F(\mathbf{X}_t, \mathbf{Z}_t) + u_{t+h}, \quad (11)$$

where $F(\cdot)$ may represent either a linear or nonlinear forecasting system. The substantive contribution of the framework lies less in the specific estimator than in the construction

of the persistence feature space. The feature vector includes several dimensions of persistence behavior:

$$\mathbf{Z}_t = \left(\hat{d}_t, \Delta \hat{d}_t, \text{Vol}(\hat{d})_t, \text{Trend}(\hat{d})_t, H_t, \Delta H_t, \bar{d}_t, \sigma_d^t, \bar{d}_{s(i),t}, \hat{d}_t \cdot \text{VIX}_t, \hat{d}_t \cdot \text{MOVE}_t \right). \quad (12)$$

These variables summarize persistence dynamics at several levels. The own-stock persistence estimate $\hat{d}_t$ captures local persistence behavior. Cross-sectional aggregates $\bar{d}_t$ and σ_d^t summarize system-wide persistence conditions, while sectoral averages $\bar{d}_{s(i),t}$ capture industry-specific uncertainty propagation. Interaction terms such as $\hat{d}_t \cdot \text{VIX}_t$ allow the impact of stress to depend on the persistence state of the market.

The forecasting architecture is organized as a layered ladder of models designed to isolate the marginal contribution of different persistence components. The baseline uses HAR components together with lagged-return controls:

$$\mathbf{Model\ A:} \quad \log \text{RV}_t^{(d)}, \log \text{RV}_t^{(w)}, \log \text{RV}_t^{(m)}, r_{t-1}, |r_{t-1}|.$$

The first augmentation introduces market-state variables:

$$\mathbf{Model\ A_1:} \quad A + (\text{VIX}_t, \text{MOVE}_t).$$

This HAR-X specification provides the market-state benchmark against which the added persistence feature set must be evaluated.

The second augmentation introduces own-stock persistence and roughness features:

$$\mathbf{Model\ A_2:} \quad A + (\hat{d}_t, \Delta \hat{d}_t, \text{Vol}(\hat{d})_t, \text{Trend}(\hat{d})_t, H_t, \Delta H_t).$$

The third and fourth augmentations introduce cross-sectional and sectoral persistence features, respectively:

$$\mathbf{Model\ A_3:} \quad A + (\bar{d}_t, \sigma_d^t),$$

$$\mathbf{Model\ A_4:} \quad A + \bar{d}_{s(i),t}.$$

The fifth augmentation introduces persistence-by-stress interactions:

$$\mathbf{Model\ A_5:} \quad A + (\hat{d}_t, \text{VIX}_t, \text{MOVE}_t, \hat{d}_t \cdot \text{VIX}_t, \hat{d}_t \cdot \text{MOVE}_t).$$

The full persistence-augmented forecasting system combines all feature blocks simultaneously, adding the persistence vector $\mathbf{Z}_t$ to HAR-X:

$$\mathbf{Model\ C:} \quad A_1 + \mathbf{Z}_t.$$

Finally, the same feature space is estimated using several shrinkage and tree-based procedures, namely the Lasso (Tibshirani, 1996), Ridge regression (Hoerl & Kennard, 1970), the Elastic Net (Zou & Hastie, 2005), Random Forests (Breiman, 2001), and Gradient Boosting (Friedman, 2001):

Model D: Lasso, Ridge, Elastic Net, Random Forest, Gradient Boosting.

These specifications are parallel augmentations of HAR, not a sequence in which each model adds to its immediate predecessor. Comparisons of A_2 , A_3 , and A_4 with HAR assess the corresponding blocks in isolation. Because A_5 includes VIX and MOVE as well

as interactions, its gain over HAR cannot be attributed to interactions alone. Model C combines the feature blocks without duplicating shared predictors; its comparison with HAR-X assesses their joint incremental contribution conditional on market-state controls.

Because Models A_2 – A_5 add persistence blocks to HAR rather than to HAR-X, their comparison with HAR-X is indirect. Three further specifications, fixed before the corresponding results were computed, add one persistence block to HAR-X directly:

$$\begin{aligned}\text{Model } A_1^{cs} : & A_1 + (\bar{d}_t, \sigma_d^t), \\ \text{Model } A_1^{sec} : & A_1 + \bar{d}_{s(i),t}, \\ \text{Model } A_1^{mod} : & A_1 + \bar{d}_t + \bar{d}_t \cdot (\log RV_t^{(d)}, \log RV_t^{(w)}, \log RV_t^{(m)}).\end{aligned}$$

The last specification lets the persistence state modulate the weights on the HAR components, which is the form the duration interpretation of Section 5 implies: if shocks decay more slowly when persistence is high, the weight on the monthly component should rise with $\bar{d}_t$.

The common feature space for Models C and D also permits comparison of linear, shrinkage, and tree-based estimators. This comparison evaluates the implemented algorithms and tuning procedures in this dataset rather than isolating a universal effect of estimator complexity.

Several implementation safeguards are required because persistence-based variables are estimated quantities rather than directly observed variables. Rolling estimation windows are constructed to avoid look-ahead bias and to preserve the forecasting information set. Feature standardization is performed using training-sample information only. Hyperparameters are selected through time-series cross-validation rather than random cross-sectional shuffling. These precautions are particularly important because volatility forecasting systems are highly sensitive to temporal dependence structures.

Forecast evaluation also requires careful treatment because volatility proxies themselves contain measurement noise. In addition to mean squared error on the logarithmic realized-volatility scale, the framework therefore employs the QLIKE loss function discussed by Patton (2011), which is more robust when realized-volatility proxies contain observational error. For pairwise forecast comparisons, the paper uses the Diebold–Mariano statistic of Diebold and Mariano (1995) with the Harvey et al. (1997) finite-sample correction (HLN), computed on the cross-sectional mean loss differential per date and using a Newey–West HAC variance (Newey & West, 1987) with bandwidth $\lceil h/5 \rceil - 1$. Importantly, inference is conducted using panel-aware cross-sectional aggregation rather than naive pooled-cell independence assumptions. This distinction is critical because ignoring within-date cross-sectional dependence substantially inflates apparent statistical significance.

Comparisons with HAR-X involve nested models, for which the Diebold–Mariano statistic is conservative: under the null hypothesis the larger model’s additional coefficients are estimation noise that raises its out-of-sample loss. We therefore also report the MSPE-adjusted statistic of Clark and West (2007), computed on the same cross-sectional mean differential with the same HAC bandwidth and compared with standard normal critical values (one-sided), and we adjust the p -values of the pre-specified HAR-X comparisons for multiplicity with the step-down procedure of Holm (1979). A specification is said to add information beyond HAR-X only if its adjusted Clark–West p -value is below 0.05 and its pooled MSE is lower than that of HAR-X; the Diebold–Mariano statistic against HAR-X is reported as the stricter criterion. Differences in Sharpe ratios in Section 9 are tested with the HAC procedure of Ledoit and Wolf (2008) and a studentized circular block bootstrap.

The forecasting structure assesses whether estimated persistence states provide useful conditional information. Horizon and regime comparisons describe where that information is most useful; they do not directly test uncertainty duration or identify the mechanisms behind persistence.

7. Data and Empirical Design

This section describes the data architecture, volatility construction procedures, rolling estimation framework, and forecasting protocol used in the empirical analysis. The central methodological principle is that persistence and roughness measures must be estimated in a manner consistent with the temporal structure of financial data while avoiding forward-looking contamination and preserving comparability across forecasting models.

The empirical framework is designed around three objectives. First, the data environment must be sufficiently long to capture multiple volatility regimes, including calm periods, financial crises, and post-crisis transitions. Second, the volatility proxies must be constructed carefully enough to distinguish genuine persistence from measurement noise. Third, the forecasting design must permit rigorous out-of-sample evaluation under realistic financial forecasting conditions.

7.1. Sample

The primary dataset consists of common equities drawn from the S&P 500 index with sufficiently long daily histories to support rolling estimation of persistence and roughness measures. Daily open, high, low, close, volume, and market-capitalization data are obtained from Bloomberg for an initial universe of 125 securities. After imposing a 70% data-coverage requirement over the sample period, the final analysis panel contains 115 stocks observed over 6,136 trading days from November 29, 2001 through April 21, 2026.¹

The 125-security universe is the S&P 500 membership at the time of the Bloomberg pull (April 2026). Index membership is not reconstructed point-in-time, and no stock enters or leaves the panel over the sample, so the sample is a fixed cross-section of firms that were constituents in 2026 and had sufficiently long histories. Requiring long histories favors surviving, established firms and excludes shorter-lived or less-completely observed securities. The results are therefore conditional on the selected panel rather than representative of all firms that belonged to the index during 2001–2026, and the coverage threshold alone does not guarantee that every retained stock is observed on every date. This selection determines which stocks are forecast; it does not change the information available at any forecast origin, because every forecast below is per-stock and walk-forward.

The sample spans several major financial regimes, including the post-2001 recovery period, the credit expansion preceding the global financial crisis, the global financial crisis itself, the European sovereign-debt episode, the long low-volatility post-crisis expansion, the COVID shock, and the post-COVID inflationary environment. The presence of these distinct regimes is essential because the paper interprets persistence measures as state-dependent indicators of uncertainty propagation.

In addition to stock-level data, the analysis incorporates several market-wide variables intended to capture broad financial conditions. These include the S&P 500 index, Dow Jones Industrial Average, Nasdaq 100, Russell 2000, the VIX implied-volatility index, the MOVE Treasury-volatility index, Treasury-yield series, the 2s/10s yield-curve slope, and investment-grade and high-yield CDX credit indices. Sector-level aggregation is implemented using GICS level-1 classifications together with SPDR sector ETFs.

¹ Data accessed through Bloomberg Terminal on April 21, 2026.

The resulting dataset provides several important advantages. The long sample horizon allows persistence measures to be evaluated across multiple crisis periods and volatility states. The cross-sectional structure permits construction of aggregate persistence measures and sectoral persistence indices.

7.2. Volatility Proxies

Daily logarithmic returns are computed from adjusted closing prices and winsorized at the 0.1% and 99.9% quantiles of each stock's own history up to and including the current day (an expanding window, applied once 500 observations are available), so that the clipping thresholds use no future information. Winsorization reduces the influence of recording errors, corporate actions, and isolated outlier observations in the predictors; the portfolio exercise of Section 9 uses raw returns. Because volatility measurement plays a central role in persistence estimation, the choice of volatility proxy is particularly important.

The primary volatility measure used throughout the paper is the range-based variance estimator of Parkinson (1980),

$$\text{RVPK}_t = \frac{(\ln H_t - \ln L_t)^2}{4 \ln 2}. \quad (13)$$

In Equation (13), H_t and L_t denote the daily high and low prices; elsewhere, H denotes the Hurst exponent. Unlike squared close-to-close returns, the Parkinson estimator exploits intraday price range information and therefore provides a more efficient volatility proxy under continuous-diffusion assumptions. The Parkinson measure offers several advantages for the present application. First, it can be constructed over the long daily sample without requiring transaction-level data. Second, it substantially reduces measurement noise relative to squared returns. Third, the estimator captures volatility clustering particularly clearly, which is important for persistence estimation.

For robustness, the forecasting comparison in Section 8 is also repeated with the logarithm of future mean squared returns as the target, a noisier close-to-close volatility proxy.

For forecasting purposes, the volatility target for stock i at horizon h is defined as the logarithm of the future average Parkinson variance,

$$y_{i,t,h} = \log\left(\frac{1}{h} \sum_{j=1}^h \text{RVPK}_{i,t+j}\right). \quad (14)$$

Three forecast horizons are considered, $h \in \{1, 5, 22\}$, corresponding approximately to daily, weekly, and monthly horizons. This design assesses whether the incremental contribution of persistence varies with the forecast horizon rather than assuming a uniform benefit.

7.3. Descriptive Statistics

Table 1 reports summary statistics for daily log returns pooled across the 115-stock panel and reported separately by GICS sector. The pooled annualized mean return equals approximately 9.3%, while the pooled annualized standard deviation equals approximately 31.7%.

More importantly, the distribution of returns displays substantial non-normality. The pooled excess kurtosis exceeds 17, and the Jarque–Bera statistic (Jarque & Bera, 1980) overwhelmingly rejects Gaussianity. The fitted Student- t distribution has approximately 2.6 degrees of freedom, indicating heavy tails. Substantial cross-sectional heterogeneity is also present across sectors. Information Technology and Financials display the largest annualized volatility levels, whereas Utilities and Consumer Staples display substantially lower

volatility. Real Estate and Financials exhibit especially heavy-tailed behavior, consistent with their sensitivity to systemic financial conditions and leverage dynamics.

Table 2 summarizes the composition of the panel together with the distribution of market-wide state variables. The VIX ranges from approximately 9 to above 82, while the MOVE index ranges from approximately 37 to above 264. These substantial variations in market stress permit evaluation of whether persistence states and their forecasting usefulness vary across market conditions.

The descriptive evidence therefore already suggests several key themes that are explored later in the forecasting analysis. Financial returns are highly non-Gaussian, volatility dynamics exhibit substantial persistence, and market-wide stress conditions vary dramatically across the sample period.

7.4. Preliminary Diagnostics

Before proceeding to rolling persistence estimation and forecasting analysis, it is important to examine several basic empirical properties of the data. The preliminary diagnostics illustrate the stylized facts motivating the paper: volatility clustering, heavy tails, cross-sectional comovement, and persistent conditional heteroskedasticity.

Figure 1 presents four preliminary diagnostics for a representative stock, together with the VIX index. Panel (a) displays daily logarithmic returns for AAPL, while panel (b) reports the corresponding Parkinson range-based variance proxy. The contrast between the two series immediately reveals one of the classical stylized facts of financial econometrics. Returns themselves fluctuate around zero with little visually detectable serial dependence, whereas volatility measures display pronounced clustering. Large volatility episodes tend to be followed by additional large volatility episodes, while calm periods tend to persist as well.

The recession-shaded regions corresponding to the global financial crisis and COVID are especially informative. During these periods, the Parkinson variance proxy rises sharply and remains elevated for extended intervals rather than reverting immediately to calm-state levels. This behavior motivates studying dependence alongside volatility magnitude, without equating a persistence estimate with a direct measure of stress duration.

Panel (c) of Figure 1 compares autocorrelation functions for raw returns, squared returns, and Parkinson realized variance. Raw returns exhibit little serial dependence at the displayed lags, consistent with the approximate unpredictability of returns documented throughout the financial-econometrics literature. By contrast, both squared returns and the Parkinson variance proxy display autocorrelations that decay slowly and remain positive beyond the first few lags. The Parkinson autocorrelations generally lie above those for squared returns, suggesting that the range-based estimator provides a less noisy volatility proxy.

This distinction is important because persistence estimation is highly sensitive to measurement error. Squared returns contain substantial noise arising from isolated jumps and market microstructure effects. By exploiting intraday high–low information, the Parkinson estimator substantially improves the signal-to-noise ratio of volatility measurement.

Panel (d) presents the VIX time series over the full sample period. The VIX displays several dramatic spikes corresponding to major stress episodes, particularly the global financial crisis and COVID. These broad market stress indicators later play a central role in the persistence-by-stress interaction framework developed in Section 6.

Figure 2 extends the diagnostic analysis in several directions. Panel (a) compares the empirical pooled distribution of daily returns with fitted Gaussian and Student-*t* distributions. The empirical distribution displays substantially heavier tails than the Gaus-

Table 1. Summary Statistics for Daily Stock Returns

	Mean (% ann.)	Std (% ann.)	Skewness	Kurtosis	Min (%)	Max (%)
Panel A: Pooled Sample						
All Stocks (N=115)	9.33	31.74	-0.21	17.35	-49.15	44.70
Panel B: By GICS Sector						
Communication Services (n=5)	12.55	33.28	-0.07	8.16	-17.06	16.94
Consumer Discretionary (n=9)	12.85	30.39	0.02	6.82	-13.79	14.08
Consumer Staples (n=11)	8.18	22.94	-0.21	8.00	-12.61	10.34
Energy (n=5)	8.61	30.68	-0.23	5.61	-13.51	12.66
Financials (n=19)	6.79	34.76	-0.14	16.68	-20.92	21.14
Health Care (n=15)	8.88	28.15	-0.34	9.76	-16.10	11.54
Industrials (n=17)	9.29	27.23	-0.27	6.16	-13.04	9.97
Information Technology (n=18)	11.20	36.55	-0.18	8.02	-20.59	16.80
Materials (n=3)	9.61	33.28	-0.28	5.23	-13.41	12.23
Real Estate (n=7)	11.55	34.84	-0.07	18.87	-25.43	22.22
Utilities (n=6)	6.40	22.60	-0.18	11.38	-13.21	12.31

Notes: The sample covers November 2001–April 2026 and includes 115 equities drawn from the S&P 500 that pass the 70% data-coverage filter. Mean returns and standard deviations are annualized; minimum and maximum returns are daily values. Kurtosis is reported as excess kurtosis. Log returns are winsorized at the 0.1% and 99.9% percentiles.

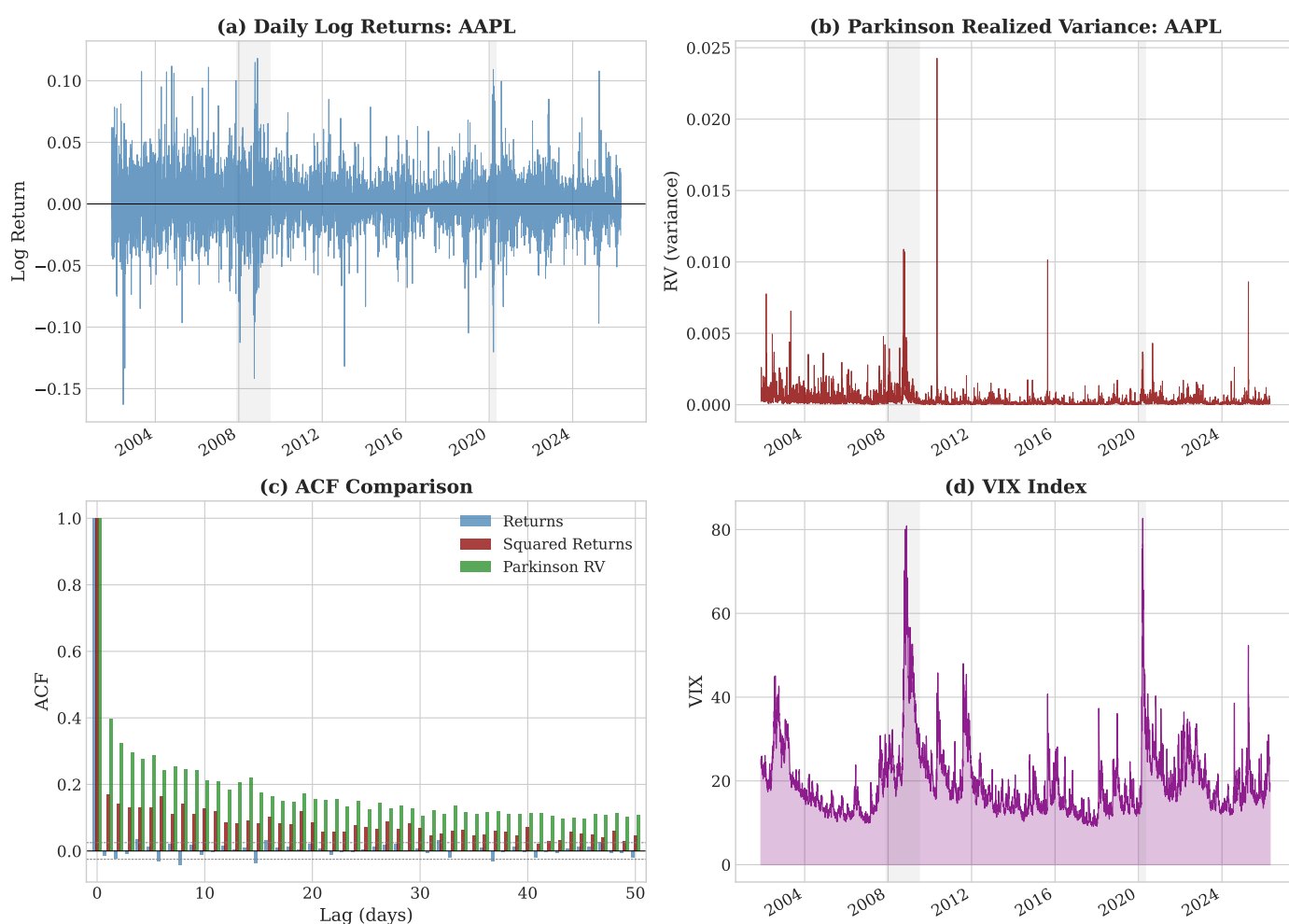


Figure 1. Daily log returns and Parkinson range-based variance proxy for a representative stock; ACF comparison; and the VIX time series. Recession-shaded windows mark the global financial crisis and COVID. *Takeaway:* volatility clusters visibly; squared returns and RVPK exhibit slow autocorrelation decay out to 50 lags while raw returns do not.

Table 2. Data Description

Panel A: Sample Composition by GICS Sector					
Sector	N Stocks	% of Sample			
Financials	19	16.5%			
Information Technology	18	15.7%			
Industrials	17	14.8%			
Health Care	15	13.0%			
Consumer Staples	11	9.6%			
Consumer Discretionary	9	7.8%			
Real Estate	7	6.1%			
Utilities	6	5.2%			
Communication Services	5	4.3%			
Energy	5	4.3%			
Materials	3	2.6%			
Total	115	100.0%			

Panel B: Market-Level Variables					
Variable	Mean	Std	Min	Max	N
SPX	2398.64	1555.37	676.53	7126.06	6136
INDU	19990.01	11002.39	6547.05	50188.14	6136
NDX	6405.01	6378.05	804.64	26672.43	6136
RTY	1183.03	601.33	327.09	2792.96	6136
VIX	19.48	8.45	9.14	82.69	6136
MOVE	88.44	31.15	36.62	264.60	6136
USGG3M	1.71	1.83	-0.14	5.51	6136
USGG10YR	3.15	1.16	0.51	5.43	6136
USYC2Y10	110.42	96.62	-108.71	291.03	6136
CDX IG 5Y	69.62	19.05	43.75	151.80	3674
CDX HY 5Y	104.59	4.19	86.25	110.41	3669

Notes: The sample covers November 2001–April 2026. Market-level variables are sourced from Bloomberg.

sian benchmark, particularly in the extremes. The associated Jarque–Bera statistic rejects normality, while the quantile–quantile (QQ) plot in panel (d) shows departures from the normal reference line in the tails. Extreme observations occur far more frequently than predicted by normal-diffusion models. This heavy-tailed structure has important implications for volatility modeling because persistence estimation can be distorted when tail behavior is ignored.

Panel (b) of Figure 2 reports cross-stock Parkinson variance correlations for ten representative equities. The displayed off-diagonal correlations range from 0.14 (PG–JPM) to 0.72 (KO–XOM). The heatmap reveals substantial cross-sectional heterogeneity in volatility comovement, with particularly large correlations among firms exposed to similar macroeconomic or sectoral conditions. These cross-sectional relationships motivate the construction of aggregate persistence variables later in the forecasting framework. If uncertainty propagates systemically across sectors and balance sheets, then cross-sectional persistence aggregates should contain information beyond individual-stock volatility dynamics.

Panel (c) compares FIGARCH conditional volatility estimates with rolling Parkinson realized volatility for AAPL. Both series respond to major stress episodes, although their short-run fluctuations differ. This is a descriptive comparison of the fractional-volatility model and the observed proxy, not a formal validation of the model. Table 3 gives FIGARCH point estimates for ten stocks with normal innovations: the mean estimate is $\hat{d} = 0.336$ with values in $[0.238, 0.480]$, all below $\frac{1}{2}$. These FIGARCH parameters are

model-specific and should not be equated with the semiparametric memory estimates or treated as a stationarity test.

Taken together, the preliminary diagnostics establish several important empirical conclusions. First, volatility measures exhibit strong persistence while returns themselves remain approximately serially uncorrelated. Second, financial returns are substantially heavy-tailed and strongly non-Gaussian. Third, volatility dynamics display substantial cross-sectional comovement. Fourth, long-memory volatility models provide empirically plausible descriptions of observed volatility behavior. Most importantly, the diagnostics strongly support the interpretation that volatility persistence is not an isolated feature of a few securities or crisis episodes. Rather, persistent volatility dynamics appear broadly throughout the cross-section and evolve systematically across market states.

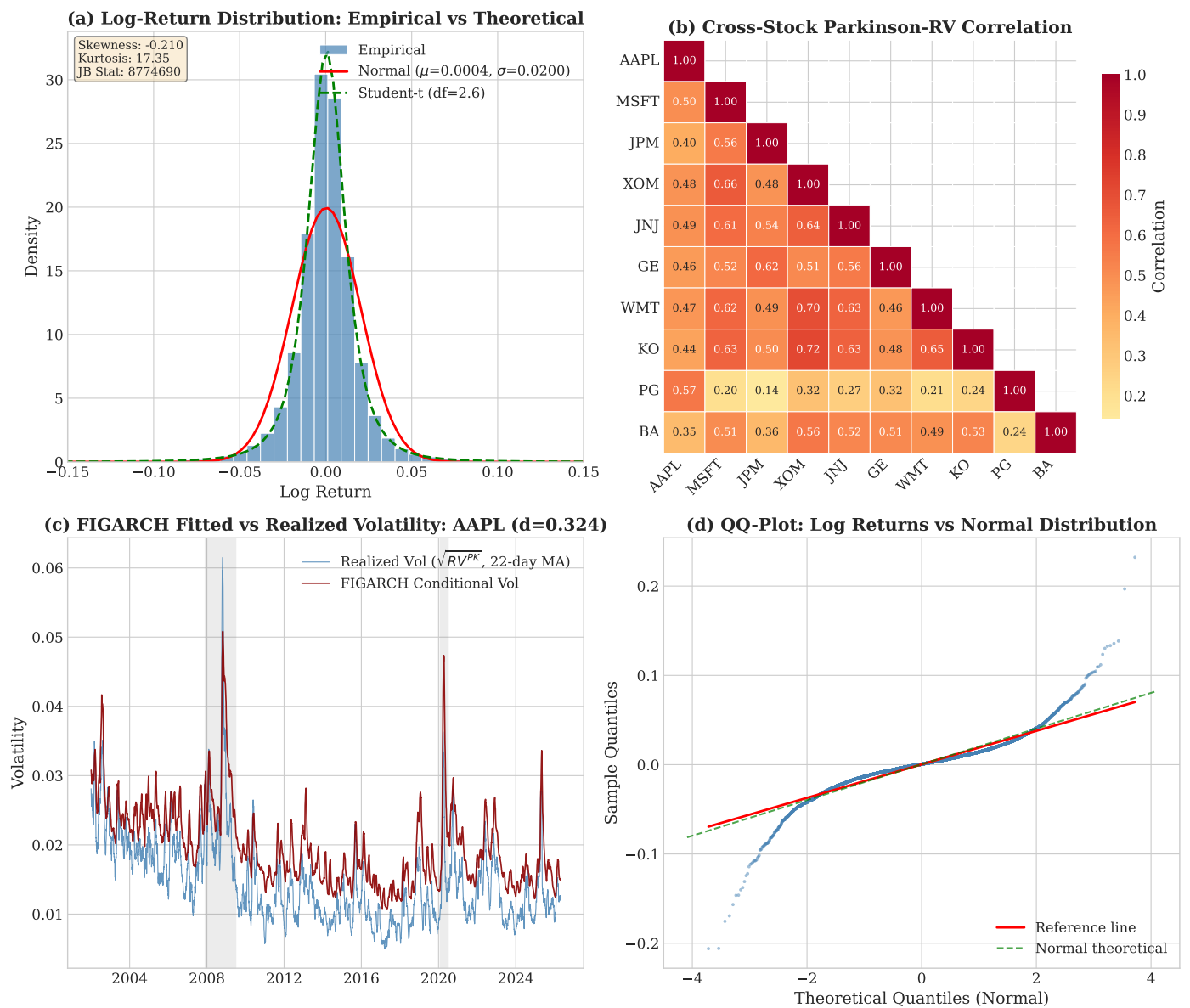


Figure 2. Preliminary diagnostics. (a) Pooled log-return distribution with normal and Student-*t* fits. (b) Cross-stock Parkinson-RV correlation heatmap. (c) FIGARCH(1,*d*,1) conditional volatility versus realized volatility for AAPL. (d) QQ-plot against a normal reference. *Takeaway:* returns have heavy tails; FIGARCH and the range-based volatility proxy both respond to major stress episodes.

Table 3. FIGARCH(1, d ,1) Model Estimates

Stock	ω	d	ϕ	β	AIC	BIC
AAPL	0.2011	0.324	0.316	0.541	25303.1	25343.5
MSFT	0.2137	0.318	0.212	0.432	22708.8	22749.2
JPM	0.1182	0.480	0.193	0.533	23456.4	23496.7
XOM	0.0682	0.416	0.230	0.551	21304.4	21344.8
JNJ	0.0706	0.327	0.336	0.496	17267.1	17307.4
GE	0.0896	0.375	0.312	0.553	23468.2	23508.6
WMT	0.1545	0.244	0.378	0.488	19536.9	19577.2
KO	0.1370	0.283	0.000	0.185	17694.4	17734.8
PG	0.1879	0.238	0.000	0.120	17626.5	17666.9
BA	0.1542	0.356	0.322	0.584	24535.3	24575.6
Mean	–	0.336	–	–	–	–

Notes: FIGARCH(1, d ,1) estimates use an AR(1) conditional mean and normal innovations. These model-specific d estimates are not directly equivalent to semiparametric memory estimates; values below 0.5 are not, by themselves, a stationarity test.

7.5. Rolling Estimation Design

Persistence and roughness measures are estimated within rolling windows so that they behave as dynamic state variables rather than fixed full-sample constants. This distinction is essential for the interpretation developed throughout the paper. If persistence reflects evolving uncertainty conditions, information diffusion, and regime-dependent stress propagation, then persistence measures themselves must be allowed to vary over time.

For each stock i and each end date t , the fractional integration parameter $\hat{d}_{i,t}$ is estimated using a rolling 750-trading-day window. Both the Geweke–Porter–Hudak log-periodogram estimator and the local-Whittle semiparametric estimator are employed using bandwidth $m = \lfloor T^{0.65} \rfloor$. The rolling estimates are updated every five trading days, producing approximately weekly time series of $\hat{d}_{i,t}^{GPH}$, $\hat{d}_{i,t}^{LW}$, and $H_{i,t}$, where $H_{i,t}$ denotes the rolling Hurst exponent estimated from logarithmic Parkinson range-based variance.

Weekly updating balances responsiveness to market conditions against short-term estimation variation. It provides an intermediate frequency between daily updates and the greater smoothing associated with monthly updates.

The 750-day estimation window is similarly motivated by the tradeoff between statistical precision and state sensitivity. Persistence estimation requires relatively long windows because long-memory parameters are fundamentally low-frequency objects. Very short windows generate unstable estimates with large sampling variance. Extremely long windows, however, blur structural regime changes and reduce responsiveness to evolving market conditions. The selected window length therefore balances these competing considerations.

The rolling framework transforms persistence from a static full-sample coefficient into a genuinely time-varying market-state variable. This transformation is central to the economic interpretation of the paper. Under the rolling design, persistence measures may rise during prolonged stress episodes and decline during calm periods, allowing analysis of state-dependent persistence, not direct measurement of stress duration.

These rolling estimates provide the forecasting features introduced in Section 6, as well as additional descriptive diagnostics. The resulting variables fall into several categories. The first category consists of own-stock persistence dynamics,

$$\hat{d}_{i,t}, \Delta \hat{d}_{i,t}, \text{Vol}(\hat{d})_{i,t}, \text{Trend}(\hat{d})_{i,t}, H_{i,t}, \Delta H_{i,t}, \quad (15)$$

summarizing the level, evolution, variability, and trend structure of persistence and roughness at the individual-stock level. The second category consists of cross-sectional persistence aggregates,

$$\bar{d}_t, \sigma_d^t, \text{Skew}_t(d), \text{Kurt}_t(d), \quad (16)$$

which summarize system-wide persistence conditions. The cross-sectional mean $\bar{d}_t$ is especially important because the paper interprets it as a summary of shared volatility dependence in financial markets. The third category consists of sectoral persistence measures, $\bar{d}_{s(i),t}$, where the average is computed across stocks belonging to the same GICS sector as stock i , intended to capture industry-specific stress propagation and heterogeneous information environments.

Additional descriptive variables include threshold indicators $\mathbf{1}_{\{\hat{d}_{i,t} > \tau\}}$ together with interaction terms involving VIX, MOVE, and liquidity conditions. The forecasting models use the predictors explicitly listed in Section 6; the broader descriptive inventory should not be interpreted as additional regressors in every model.

Table 4 reports pooled descriptive statistics for the resulting feature set. Several patterns are immediately noteworthy. First, the GPH and local-Whittle rolling estimates are close on average (pooled means of 0.346 and 0.336), although the GPH estimates are somewhat more dispersed. Second, the reported Hurst estimates are well below $\frac{1}{2}$, consistent with roughness in the observed volatility proxy. Third, the cross-sectional mean persistence measure varies substantially over time, supporting its interpretation as a dynamic state variable rather than a stable structural constant.

The rolling estimates are not uniformly within the covariance-stationary fractional range: the reported 99th percentiles are 0.8308 (GPH) and 0.8278 (local-Whittle), above $d = \frac{1}{2}$, and roughly one fifth of the rolling GPH estimates exceed that threshold. A mean estimate below the threshold does not establish stationarity in every window. We therefore use rolling estimates as reduced-form dependence features and do not interpret large values as verified parameters of a stationary fractional process.

The persistence-by-VIX interaction is larger during crisis periods. Because it is a product of its inputs, this descriptive pattern alone does not establish an amplification mechanism.

Rolling persistence features are constructed from observations available through date t , preserving the forecasting information set. The walk-forward protocol is described below.

The rolling framework permits comparison of estimated dependence across market states. Because long windows smooth and overlap, movements in these estimates should not be read as dates of shock arrival or measurements of how quickly a particular disturbance is resolved.

Moreover, because the rolling framework generates persistence measures for each stock, sector, and date simultaneously, it becomes possible to study not only average persistence but also the cross-sectional dispersion and heterogeneity of persistence conditions throughout the financial system. The resulting persistence-state panel therefore forms the core empirical object underlying the forecasting analysis developed in the next section.

7.6. Out-of-Sample Forecasting Protocol

The forecast evaluation is strictly walk-forward and point-in-time. We use the first 40% of the sample dates (sample dates 1 through 431, ending in June 2013) as the initial training window; thereafter, we produce forecasts at every subsequent sample date. At each origin t the training sample contains only rows whose target window ends on or before the origin day, that is, rows j with $\text{pos}_j + h \leq \text{pos}_t$ in trading days. With a five-day sampling stride this excludes no rows at $h = 1$ or $h = 5$ (the weekly target of the

Table 4. Persistence Features and Descriptive Diagnostics: Pooled Statistics

Category	Feature	Mean	Std	1%	99%
LRD					
	$\hat{d}_{GPH,t}$ (rolling)	0.3461	0.1922	0.0016	0.8308
	$\hat{d}_{LW,t}$ (rolling)	0.3358	0.1738	0.0265	0.8278
Roughness					
	H_t (rolling)	0.0620	0.0275	0.0196	0.1303
Memory dyn.					
	$\Delta \hat{d}_t$	-0.0002	0.0300	-0.0956	0.0956
	$\text{Vol}(\hat{d})_t$	0.0134	0.0224	0.0001	0.1174
	$\text{Trend}(\hat{d})_t$	-0.0002	0.0161	-0.0532	0.0504
HAR					
	RV^d	0.0003	0.0012	0.0000	0.0027
	RV^w	0.0003	0.0010	0.0000	0.0027
	RV^m	0.0003	0.0008	0.0000	0.0025
Sector aggregate					
	$\hat{d}_{s(i),t}$	0.3461	0.1605	0.0891	0.6724
Cross-sectional					
	$\bar{d}_t$ (cross-sectional mean)	0.3461	0.1493	0.1400	0.6058
	σ_d^t (cross-sectional std)	0.1192	0.0236	0.0670	0.1676
	Skew of d_t across stocks	0.0165	0.4559	-1.1219	0.8546
	Kurt of d_t across stocks	0.3385	0.8753	-0.7179	2.8986
	$P(\hat{d} > 0.30)$	0.5311	0.3313	0.0522	1.0000
	Range of $\hat{d}$ across stocks	0.6155	0.1391	0.3810	0.9985
Market					
	VIX	19.0494	8.3693	9.8977	53.7306
	MOVE	84.6779	30.7937	44.3771	188.2890
Interaction					
	$\hat{d}_t \cdot \text{VIX}_t$	7.2215	6.2893	0.0250	29.4611
	$\hat{d}_t \cdot \text{MOVE}_t$	30.6675	23.2602	0.1071	108.2395

Notes: Statistics are pooled across stocks and dates in the rolling-estimation panel. Persistence estimates are updated every five trading days, and HAR components are constructed from Parkinson range-based variance. Sector aggregates use GICS level-1 classifications. Threshold indicators and interactions with VIX and MOVE use GPH estimates. This table includes descriptive diagnostics in addition to the forecasting predictors specified in Section 6.

preceding row ends exactly on the origin day) and the four most recent rows at $h = 22$, whose monthly windows would otherwise extend up to 17 trading days past the origin. All predictors dated t are known at the close of day t : the HAR components and the rolling persistence and roughness estimates end on day $t - 1$, the last completed observation before the origin, while VIX and MOVE are the closes on day t . For the linear specifications we refit by ordinary least squares at every step (expanding window). For Model D we refit every twenty sample steps to keep the computational burden tractable for the tree-based estimators, applying the same exclusion rule at each refit; results are not sensitive to this stride. All feature standardization, hyperparameter selection, and model selection are performed on training-window information only. Both the realized target and the forecast are recorded as $(T_{\text{eval}} \times N)$ panels, which are then used to calculate forecast losses, the test statistics, and the regime and sector comparisons in Section 8.

8. Numerical Analysis and Forecasting Results

This section presents the empirical results of the paper. The analysis proceeds in several stages. First, we examine full-sample estimates of long-memory and roughness parameters across the cross-section of equities. Second, we study the dynamics of rolling persistence measures across market regimes. Third, we evaluate the forecasting performance of the layered forecasting framework developed in Section 6. Finally, we investigate robustness across sectors, volatility states, liquidity conditions, and alternative estimation procedures.

The central empirical question is whether persistence measures behave like economically meaningful state variables rather than purely statistical nuisance parameters. Under the interpretation developed earlier, persistence measures should rise during sustained stress episodes, co-move with broad uncertainty indicators, and contribute forecasting information particularly during periods in which uncertainty resolves slowly.

8.1. Long-Memory and Roughness Estimates

We begin with full-sample cross-sectional estimates of persistence and roughness. The results indicate persistent dependence and low Hurst estimates in the observed volatility proxies across the equity panel.

The first important finding is that raw returns themselves display essentially no evidence of long memory. The cross-sectional mean GPH estimate for returns is approximately $\hat{d}_{\text{returns}}^{\text{GPH}} = -0.022$, and only 15% of stocks exhibit statistical significance at the 5% level. This result is consistent with the approximate unpredictability of financial returns documented throughout the empirical asset-pricing literature.

The volatility proxies show a different pattern. For the Parkinson range-based variance proxy, the cross-sectional mean GPH estimate equals approximately $\hat{d}_{\text{RV}}^{\text{GPH}} = 0.451$, while the corresponding local-Whittle estimate equals approximately $\hat{d}_{\text{RV}}^{\text{LW}} = 0.440$. The two semiparametric estimators therefore agree closely, both on average and sector by sector (Table 5). Moreover, 99% (GPH) and 100% (local-Whittle) of stocks exhibit statistically significant persistence estimates. These results provide strong evidence that long-memory behavior is a broad characteristic of volatility dynamics rather than an isolated feature of a few securities.

The roughness diagnostics provide complementary evidence. The cross-sectional mean Hurst exponent estimated from logarithmic Parkinson range-based variance equals approximately $H = 0.063$, with every stock in the panel satisfying $H < \frac{1}{2}$. This result is consistent with local irregularity in the observed proxy; it does not by itself identify the path regularity of latent volatility. The cross-sectional dispersion of Hurst estimates is small, indicating similar proxy-based roughness estimates across sectors and firms.

Table 5. Long-Range Dependence and Roughness Estimates by Sector

Sector	Returns		Parkinson RV			Roughness	
	$\bar{d}_{\text{GPH}}$	$\bar{d}_{\text{LW}}$	$\bar{d}_{\text{GPH}}$	$\bar{d}_{\text{LW}}$	% Sig	$\bar{H}$	N
Communication Services	0.019	0.002	0.477	0.477	100%	0.076	5
Consumer Discretionary	-0.019	-0.028	0.454	0.444	100%	0.056	9
Consumer Staples	-0.037	-0.041	0.384	0.381	91%	0.059	11
Energy	-0.008	-0.030	0.416	0.439	100%	0.067	5
Financials	-0.027	-0.033	0.490	0.461	100%	0.071	19
Health Care	-0.034	-0.030	0.431	0.419	100%	0.058	15
Industrials	-0.011	-0.015	0.444	0.437	100%	0.060	17
Information Technology	-0.015	-0.017	0.467	0.455	100%	0.061	18
Materials	-0.019	-0.017	0.464	0.448	100%	0.061	3
Real Estate	-0.011	-0.037	0.511	0.486	100%	0.069	7
Utilities	-0.063	-0.066	0.400	0.404	100%	0.062	6
Overall	-0.022	-0.028	0.451	0.440	99%	0.063	115

Notes: d denotes the fractional differencing parameter. GPH is the Geweke–Porter–Hudak log-periodogram estimator with bandwidth $m = \lfloor T^{0.65} \rfloor$; LW is the local-Whittle estimator. Returns are daily log returns; Parkinson RV is the range-based variance proxy constructed from daily high and low prices. H is the Hurst exponent estimated from the scaling of the $q = 2$ absolute moment of increments in $\log \text{RV}^{\text{PK}}$ over lags $\{1, 2, 3, 5, 8, 13, 21\}$. Estimates below $H = 0.5$ are consistent with roughness in the observed proxy. % Sig reports the share of stocks whose volatility GPH estimates are significant at the 5% level.

Table 5 reports these estimates separately by GICS sector. Financials and Real Estate have relatively large persistence estimates, consistent with their exposure to systemic stress propagation, leverage conditions, and liquidity cycles. Utilities and Consumer Staples display somewhat smaller persistence estimates, although long-memory behavior remains substantial throughout all sectors. An additional feature of the results is the limited dispersion of persistence estimates across the cross-section: the standard deviation of persistence estimates across stocks remains relatively modest, indicating that long-memory volatility behavior is fundamentally a panel-wide phenomenon rather than the consequence of a few extreme outliers.

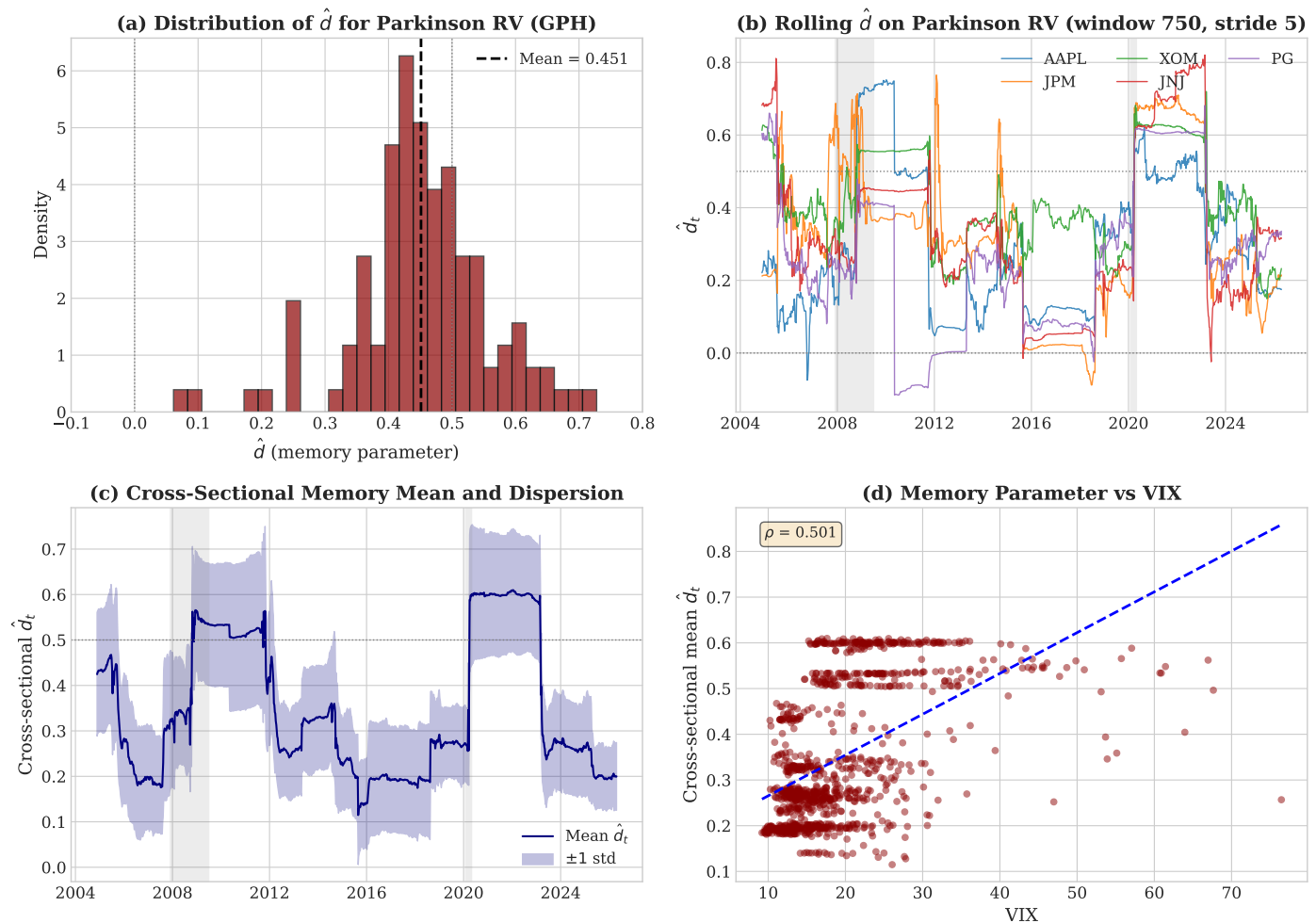


Figure 3. Rolling persistence estimates. (a) Cross-sectional distribution of full-sample GPH $\hat{d}$ on Parkinson range-based variance proxy. (b) Rolling $\hat{d}^{GPH}$ for AAPL, JPM, XOM, JNJ, PG. (c) Cross-sectional mean $\bar{d}_t$ with ± 1 standard deviation band. (d) $\bar{d}_t$ versus VIX, with linear fit and correlation. *Takeaway: the cross-sectional mean memory parameter rises sharply in the GFC and COVID and tracks the VIX with correlation $\rho = +0.50$, consistent with its interpretation as a stress-associated persistence state.*

Figure 3 reports four time-series and cross-sectional views of the rolling estimation. Panel (a) reports the distribution of full-sample GPH estimates across all 115 stocks. The distribution is concentrated between 0.30 and 0.60 (84% of stocks) around the mean of 0.451; about 30% of the full-sample estimates lie above $\frac{1}{2}$, the boundary of the covariance-stationary range, which is one reason the paper treats $\hat{d}$ as a reduced-form dependence measure rather than a verified parameter of a stationary fractional process.

Panel (b) reports rolling $\hat{d}^{GPH}$ for five representative stocks (AAPL, JPM, XOM, JNJ, PG). The most striking feature is the broad upward shift in persistence during major crisis episodes, particularly during the global financial crisis and COVID.

Panel (c) presents the aggregate state central to the forecasting framework: the cross-sectional mean persistence state $\bar{d}_t = N^{-1} \sum_i \hat{d}_{i,t}$, together with a one-standard-deviation band. The dynamics of this aggregate persistence state are highly informative. During the calm 2013–2014 period, the cross-sectional mean persistence measure is approximately $\bar{d}_t \approx 0.31$. During the global financial crisis (2008–Q3 through 2009–Q4), the cross-sectional mean rises to approximately $\bar{d}_t \approx 0.50$, an increase of approximately 62%. During COVID (March–December 2020), the persistence state rises even further to approximately $\bar{d}_t \approx 0.57$, an increase of approximately 86% relative to the calm-state baseline.

These increases document an association between estimated persistence and major stress episodes. They are consistent with prolonged uncertainty propagation, but do not

measure the time needed to resolve uncertainty or distinguish that interpretation from regime-dependent changes in volatility.

Panel (d) of Figure 3 examines the relationship between the cross-sectional persistence state and the VIX. The estimated contemporaneous correlation equals approximately $\rho(\bar{d}_t, \text{VIX}_t) = 0.50$. This correlation shows that persistence and VIX are related but not identical. It does not establish conditional forecast value, a leading relationship, or the expected duration of uncertainty. The benchmark and lead-lag comparisons below assess predictive usefulness and temporal association, not duration.

Taken together, these findings motivate the reduced-form interpretation of Section 5: persistence varies with market conditions and may summarize shared dependence relevant to forecasting. The next subsection tests its predictive usefulness beyond HAR and HAR-X rather than assuming that its association with stress establishes that usefulness.

8.2. Out-of-Sample Forecast Comparison

Table 6 compares the HAR core (Model A), its parallel augmentations (A_1 – A_5), the full feature specification (C), and alternative estimators (D) at horizons $h \in \{1, 5, 22\}$. Performance is measured by mean squared error on the log Parkinson-variance target and QLIKE on the variance scale. Positive percentage improvements denote lower MSE than HAR. The reported panel-aware HLN-corrected Diebold–Mariano statistics also use HAR as the reference; they do not test Model C against HAR-X.

The HAR-X benchmark is essential for interpreting the results. Adding VIX and MOVE alone (A_1) reduces MSE relative to HAR by 5.10%, 7.73%, and 2.82% at the daily, weekly, and monthly horizons. The reported tests reject equal predictive accuracy against HAR at the daily and weekly horizons ($t = 5.34$ and 5.31) but not at the monthly horizon ($t = 1.49$). Market-state variables therefore account for a substantial part of the improvement available over volatility lags alone at short horizons.

Adding only own-stock persistence and roughness (A_2) does not improve MSE: the corresponding changes relative to HAR are -0.54% , -0.03% , and -1.89% . The daily and monthly deteriorations are statistically significant under the reported test. This result shows that estimating individual persistence does not automatically provide information beyond the multi-horizon HAR components.

Cross-sectional aggregates alone (A_3) yield changes of -0.10% , $+0.37\%$, and $+0.05\%$, while sectoral persistence alone (A_4) yields -0.05% , $+0.21\%$, and -0.16% . None of these changes is statistically significant at the reported levels. These blocks are therefore potential components of a joint state representation, not independently established sources of forecast gains.

Model A_5 , which includes market controls as well as persistence–stress interactions, improves on HAR by 4.91%, 7.63%, and 0.39%. Each of these MSE gains is smaller than the corresponding HAR-X gain. Its performance cannot be read as evidence that interactions alone outperform market-state controls, although its QLIKE values are slightly lower than those of HAR-X at the daily and weekly horizons.

The full persistence-augmented specification (C) reduces MSE relative to HAR by 4.55%, 8.25%, and -0.62% . Its reported HLN statistics are 3.15, 3.88, and -0.21 , respectively, indicating significance against HAR at the 1% level at the daily and weekly horizons and no difference at the monthly horizon. Relative to HAR-X the pattern is stark. Model C has higher daily MSE (0.6382 versus 0.6345), slightly lower weekly MSE (0.3335 versus 0.3353), and higher monthly MSE (0.2726 versus 0.2633). Expressed on the common HAR-relative improvement scale, these differences are -0.55 , $+0.52$, and -3.44 percentage points. Direct significance tests of C and of every other specification against HAR-X are reported in Section 8.3.

Table 6. Out-of-Sample Forecast Comparison Across Model Specifications at Horizons $h \in \{1, 5, 22\}$

Model	h	MSE(log RV^{PK})	QLIKE	% Δ vs A	HLN DM- t vs A
A	1	0.6687	-7.5872	+0.00%	–
	5	0.3634	-7.6731	+0.00%	–
	22	0.2710	-7.6101	+0.00%	–
A1	1	0.6345	-7.6339	+5.10%	+5.34***
	5	0.3354	-7.6948	+7.73%	+5.31***
	22	0.2633	-7.6118	+2.82%	+1.50
A2	1	0.6723	-7.5781	-0.54%	-3.05***
	5	0.3636	-7.6712	-0.03%	-0.07
	22	0.2761	-7.6027	-1.89%	-2.86***
A3	1	0.6693	-7.5918	-0.10%	-0.59
	5	0.3621	-7.6755	+0.36%	+1.04
	22	0.2708	-7.6123	+0.05%	+0.05
A4	1	0.6690	-7.5843	-0.05%	-0.46
	5	0.3627	-7.6717	+0.21%	+0.81
	22	0.2714	-7.6078	-0.16%	-0.24
A5	1	0.6358	-7.6441	+4.91%	+4.02***
	5	0.3357	-7.6970	+7.63%	+4.18***
	22	0.2699	-7.6103	+0.39%	+0.20
C	1	0.6382	-7.6421	+4.55%	+3.15***
	5	0.3335	-7.6992	+8.25%	+3.88***
	22	0.2727	-7.6114	-0.62%	-0.22
D_lasso	1	0.6403	-7.6316	+4.24%	+3.93***
	5	0.3381	-7.6940	+6.98%	+4.16***
	22	0.2703	-7.6105	+0.23%	+0.15
D_ridge	1	0.6452	-7.6301	+3.51%	+3.28***
	5	0.3418	-7.6931	+5.96%	+3.60***
	22	0.2723	-7.6112	-0.51%	-0.15
D_en	1	0.6405	-7.6305	+4.21%	+3.98***
	5	0.3382	-7.6939	+6.94%	+4.23***
	22	0.2704	-7.6108	+0.20%	+0.13
D_rf	1	0.6658	-7.6177	+0.43%	+0.42
	5	0.3553	-7.6824	+2.23%	+1.63
	22	0.2892	-7.5983	-6.74%	-2.80***
D_gbm	1	0.7584	-7.5710	-13.42%	-8.57***
	5	0.3993	-7.6656	-9.86%	-4.26***
	22	0.3380	-7.5802	-24.76%	-6.42***

Notes: Results are pooled across stocks and out-of-sample dates following the initial 40% training window. MSE is calculated on the log RV^{PK} scale and QLIKE on the variance scale. HLN DM- t denotes the Harvey–Leybourne–Newbold finite-sample-corrected Diebold–Mariano statistic, calculated from the cross-sectional mean loss differential at each date using a Newey–West HAC variance with bandwidth $[h/5] - 1$ and a Student- $t(T - 1)$ reference distribution, where T is the number of evaluation dates. The bandwidth accounts for multi-step forecast overlap under five-trading-day sampling. Positive improvements and test statistics favor the model over A. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Model A contains HAR components and lagged-return controls. Models A₁–A₅ are parallel augmentations of A: A₁ adds VIX and MOVE (HAR-X); A₂ adds own-stock persistence and roughness; A₃ adds cross-sectional mean persistence and dispersion; A₄ adds sectoral mean persistence; and A₅ adds $\hat{d}$, VIX, MOVE, and the interactions $\hat{d} \times \text{VIX}$ and $\hat{d} \times \text{MOVE}$. Model C combines these predictors without duplication. The D specifications use C's predictors with shrinkage or tree-based estimators. All reported tests compare the models with A, not with HAR-X.

Thus, market-state variables account for essentially all of the gain over HAR, and at the monthly horizon the joint persistence set is worse than HAR-X. Under QLIKE the full model is marginally below HAR-X at the daily and weekly horizons and above it at the monthly horizon, so the ranking is not an artifact of the loss function. This evidence supports persistence at most as a complement to established predictors, not as a source of additional accuracy.

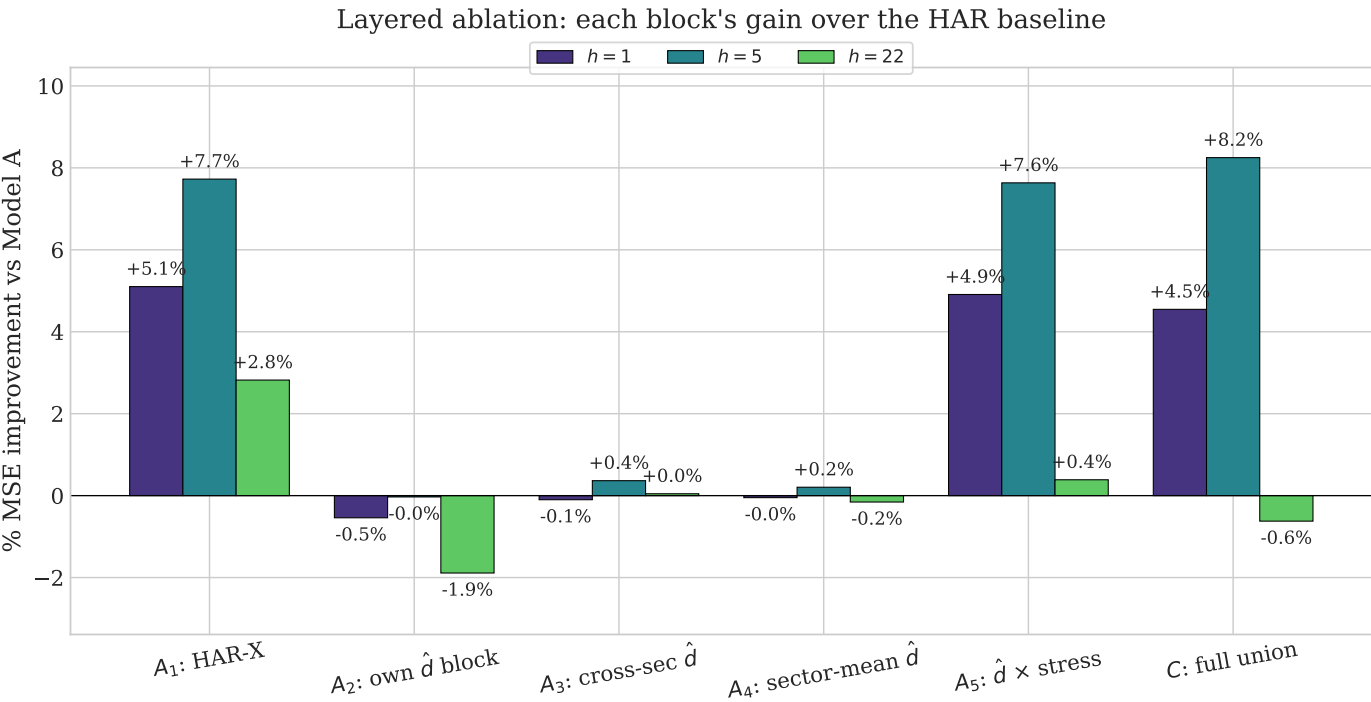


Figure 4. Forecast gains relative to HAR. Bars report percentage MSE improvement over Model A at $h \in \{1, 5, 22\}$. Models A_1 – A_5 are parallel augmentations, not sequential marginal additions.

The estimator comparisons favor the structured linear specification among models using the full feature set. Lasso, Ridge, and Elastic Net improve on HAR at daily and weekly horizons but offer little monthly improvement. Random Forest yields modest short-horizon gains and deteriorates at the monthly horizon; Gradient Boosting increases MSE at every pooled horizon. Overfitting is a possible explanation, but these comparisons do not identify its cause or establish that nonlinear methods are generally inferior. They show that the evaluated nonlinear estimators do not deliver additional accuracy from this feature set under the implemented protocol.

Figure 5 shows that the weekly gains over HAR are broadly distributed: approximately 99% of stocks improve under HAR-X and 94% under Model C. These shares concern comparisons with HAR, not the fraction of stocks benefiting incrementally from persistence beyond HAR-X. Together with Table 6, they support a narrower conclusion: own-stock persistence alone is unhelpful, and the breadth of the joint specification’s gains over HAR is the breadth of HAR-X’s gains.

8.3. Incremental Accuracy Relative to HAR-X

Table 7 tests every specification directly against HAR-X. Three facts stand out. First, by the Diebold–Mariano criterion Model C is never significantly more accurate than HAR-X: the HLN statistics are -0.74 , $+0.48$, and -2.44 at the daily, weekly, and monthly horizons. Second, the Clark–West statistic rejects the nested null for Model C at every horizon ($t = 6.1, 7.7, 6.3$) and for the pre-specified A_1^{cs} and A_1^{mod} specifications at the daily and weekly horizons (Holm-adjusted $p \leq 0.009$): the persistence regressors have non-zero

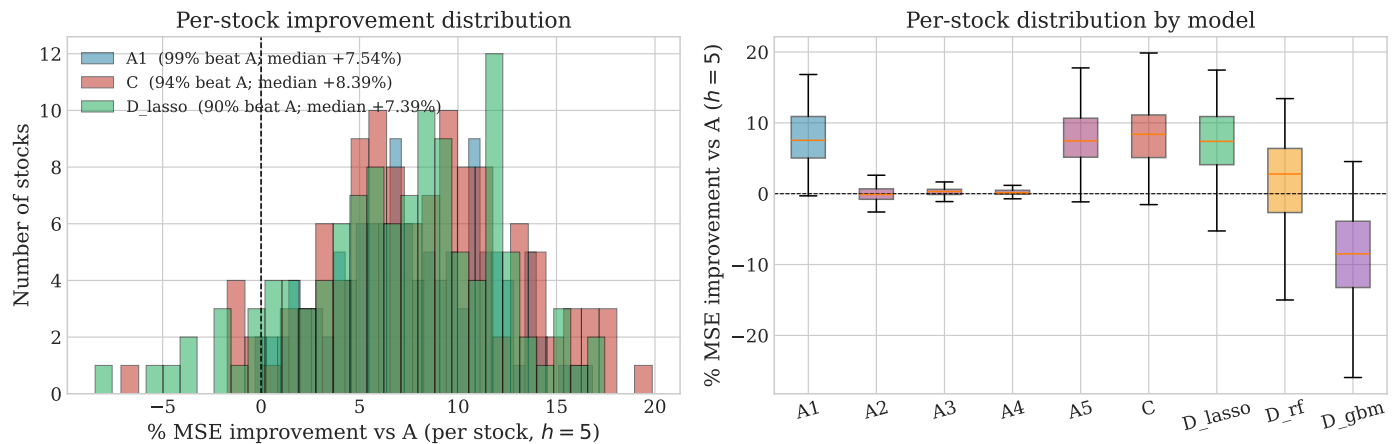
Per-stock forecast-improvement distribution (115 stocks, $h = 5$)

Figure 5. Per-stock MSE improvement versus Model A at $h = 5$. Left: histograms for A_1 , C, and D-Lasso. Right: distributions across the forecasting specifications.

Table 7. Incremental Forecast Accuracy Relative to HAR-X: Point-in-Time Out-of-Sample Tests

Specification	h	% Δ MSE vs HAR-X	HLN DM- t	CW- t	CW p (Holm)	QLIKE DM- t
HAR-X + cross-sectional $\bar{d}_t, \sigma_d^t$	1	-0.15%	-0.38	+3.21***	0.005	+1.64
	5	-0.18%	-0.37	+3.36***	0.004	+2.52**
	22	-1.66%	-1.67*	+1.08	0.409	+1.03
HAR-X + sector-mean $\bar{d}_{s(i),t}$	1	-0.19%	-0.63	+2.20**	0.069	+1.20
	5	-0.38%	-1.06	+1.96**	0.099	+1.37
	22	-1.42%	-2.62***	+0.49	0.409	+0.11
HAR-X + $\bar{d}_t \times$ HAR components	1	-0.61%	-1.12	+2.96***	0.009	+0.87
	5	-0.60%	-0.80	+3.12***	0.006	+0.93
	22	-2.25%	-2.07**	+1.10	0.409	-0.07
Model C (full union)	1	-0.58%	-0.74	+6.13***	0.000	+0.54
	5	+0.57%	+0.48	+7.75***	0.000	+1.20
	22	-3.54%	-2.44**	+6.25***	0.000	+0.30
Model D, Lasso	1	-0.91%	-2.70***	+5.68***	–	+0.07
	5	-0.81%	-1.81*	+7.58***	–	+0.04
	22	-2.68%	-2.34**	+5.00***	–	+0.18
Model D, Ridge	1	-1.68%	-4.23***	+5.52***	–	-0.20
	5	-1.92%	-3.86***	+7.78***	–	-0.52
	22	-3.42%	-2.42**	+4.68***	–	+0.30
Model D, Elastic Net	1	-0.94%	-2.85***	+5.46***	–	-0.13
	5	-0.85%	-1.99**	+7.81***	–	+0.00
	22	-2.70%	-2.27**	+4.88***	–	+0.24
Model D, Random Forest	1	-4.93%	-6.15***	+3.93***	–	-1.24
	5	-5.95%	-5.08***	+5.74***	–	-2.61***
	22	-9.83%	-4.11***	+5.62***	–	-1.74*
Model D, Gradient Boosting	1	-19.52%	-19.71***	+5.55***	–	-3.01***
	5	-19.05%	-13.29***	+7.96***	–	-6.76***
	22	-28.38%	-8.05***	+4.97***	–	-4.94***

Notes: All forecasts are point-in-time (training rows whose target window extends past the origin are excluded). % Δ MSE is the change in pooled MSE on $\log RV^{PK}$ relative to HAR-X (Model A_1); positive favours the specification. HLN DM- t : panel-aware Harvey–Leybourne–Newbold-corrected Diebold–Mariano statistic against HAR-X (two-sided). CW- t : Clark–West (2007) MSPE-adjusted statistic for the nested comparison with HAR-X (one-sided, standard normal). Both use the cross-sectional mean loss differential per date with a Newey–West bandwidth $[h/5] - 1$. CW p (Holm) adjusts the three pre-registered specifications' nine tests jointly and Model C's three tests jointly; other rows are unadjusted (–). Significance of raw statistics: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

population coefficients. Third, that information does not survive estimation. None of the three HAR-X-nesting specifications lowers pooled MSE relative to HAR-X at any horizon (the changes range from -0.15% to -2.25%), and Model C does so only at the weekly horizon and only by 0.57% . Under the decision rule stated in Section 6, Model C at $h = 5$ is the single case that qualifies, and its Diebold–Mariano statistic against HAR-X is $+0.48$. The A_1^{mod} specification, which lets the persistence state modulate the HAR weights as the duration interpretation implies, is worse than HAR-X at all three horizons.

Table 8. Pre-Registered Follow-Up Designs: Does the Persistence Information Become Usable?

Comparison	h	% Δ MSE	HLN DM- t	CW- t	CW p (Holm)
<i>Panel A: pooled estimation with stock fixed effects (level target)</i>					
P-C vs P-A1	1	+0.19%	+0.36	+2.55***	0.033
P-A1cs vs P-A1	1	-0.05%	-0.13	+1.65**	0.199
P-A1 vs per-stock A1	1	+0.06%	+0.71	+9.73***	–
P-A1 vs P-A	1	+4.80%	+5.53***	+6.74***	–
P-C vs P-A1	5	+0.19%	+0.25	+2.27**	0.058
P-A1cs vs P-A1	5	-0.36%	-0.81	+0.65	0.770
P-A1 vs per-stock A1	5	-0.72%	-2.18**	+7.05***	–
P-A1 vs P-A	5	+6.87%	+5.30***	+7.10***	–
P-C vs P-A1	22	-1.60%	-1.31	-0.10	1.000
P-A1cs vs P-A1	22	-1.57%	-1.84*	-1.11	1.000
P-A1 vs per-stock A1	22	-0.01%	+0.03	+6.47***	–
P-A1 vs P-A	22	+2.65%	+1.62	+4.72***	–
<i>Panel B: duration target $D_t = y_{t,22} - y_{t,5}$, per-stock OLS</i>					
A1 vs A (duration target)	22	+0.14%	+0.20	+2.25**	–
A1cs vs A1 (duration target)	22	-0.13%	-0.44	+1.07	0.142
C vs A1 (duration target)	22	-1.79%	-3.12***	+2.34***	0.020
<i>Panel C: market-level target (cross-sectional mean), single series</i>					
M-A1 vs M-A	1	+11.51%	+5.05***	+6.13***	–
M-C vs M-A1	1	-1.22%	-0.49	+1.72**	0.129
M-A1 vs M-A	5	+10.86%	+4.44***	+6.26***	–
M-C vs M-A1	5	-1.90%	-0.93	+1.09	0.276
M-A1 vs M-A	22	+0.73%	+0.22	+2.51***	–
M-C vs M-A1	22	-6.66%	-3.13***	-2.22	0.987

Notes: All forecasts are point-in-time expanding-window OLS. % Δ MSE is the change in pooled MSE of the second-named model relative to the first (positive favours the second). HLN DM- t (two-sided) and Clark–West CW- t (one-sided, nested) as in Table 7. Holm adjustment is applied within each panel to the pre-registered comparisons; other rows are unadjusted (–). Panel A: one regression per origin on the stacked panel of all stocks with stock fixed effects (within transformation on training rows). Panel B: the target is the log ratio of monthly to weekly future mean variance. Panel C: a single market series with HAR components of the cross-sectional mean log variance; $T \approx 645$ evaluation dates. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The combination of Clark–West rejections and flat or negative MSE differences points to estimation noise: each per-stock regression estimates between three and thirteen additional coefficients on a few hundred weekly observations. Table 8 reports three designs, fixed in advance, that address this and test the economic interpretation directly. Pooled estimation with stock fixed effects (Panel A) recovers a gain of 0.20% for the full persistence set over pooled HAR-X at the daily horizon (Clark–West Holm-adjusted $p = 0.033$, Diebold–Mariano $t = 0.36$), 0.19% at the weekly horizon ($p = 0.058$), and a loss at the monthly horizon; pooling does not improve HAR-X itself. The estimation-noise explanation is therefore correct, but the recoverable information is economically negligible. Panel B replaces the level target with the log ratio of monthly to weekly future variance, the quantity that a duration measure should predict: adding the cross-sectional persistence state to HAR-X changes MSE by -0.13% , and the full set by -1.80% (Diebold–Mariano $t = -3.12$). Panel C forecasts the cross-sectional mean of the stock targets as

a single market series: the persistence state lowers accuracy relative to market HAR-X at every horizon (-1.2% , -1.9% , -6.7%). To a close approximation, the persistence state's forecasting content is its co-movement with the VIX.

8.4. Descriptive Coefficient Diagnostics

Table 9 reports standardized Lasso coefficients from one pooled fit per horizon using all out-of-sample evaluation dates. These are retrospective descriptive fits, not the walk-forward forecasts evaluated in Table 6. They describe conditional associations within the feature set rather than providing independent out-of-sample validation or a causal decomposition of forecast gains.

Table 9. Standardized Lasso Coefficients from the Pooled Evaluation Sample

Feature	$h = 1$	$h = 5$	$h = 22$
har_d_log	+0.1337	+0.0999	+0.0684
har_w_log	+0.2106	+0.1840	+0.1460
har_m_log	+0.3539	+0.4046	+0.4497
ret_lag1	-0.0385	-0.0470	-0.0419
ret_lag1_abs	+0.0232	+0.0143	+0.0164
d_gph	+0.0878	+0.0836	+0.0527
delta_d_gph	+0.0113	+0.0070	+0.0029
vol_d_gph	-0.0252	-0.0393	-0.0465
trend_d_gph	-0.0076	-0.0077	-0.0000
h	-0.0168	-0.0091	-0.0060
delta_h	+0.0043	+0.0153	+0.0094
cs_mean_d	-0.0673	-0.0474	-0.0148
cs_std_d	-0.0067	-0.0058	-0.0222
sector_mean_d	+0.0147	+0.0085	+0.0094
vix	+0.2784	+0.2247	+0.1133
move	-0.0315	+0.0000	+0.0214
d_x_vix	-0.1452	-0.1431	-0.1150
d_x_move	+0.0228	+0.0240	+0.0382

Notes: One retrospective pooled Lasso fit is estimated per horizon using standardized predictors for all stocks and out-of-sample evaluation dates. Standardization permits comparison of coefficient magnitudes, but these fits do not provide independent out-of-sample validation or significance tests. Zero coefficients indicate predictors excluded by Lasso.

The HAR components remain prominent. The monthly component has the largest coefficient at each horizon, rising from 0.3540 at $h = 1$ to 0.4499 at $h = 22$. VIX also enters positively, with a coefficient declining from 0.2784 to 0.1132. These patterns reinforce the importance of established volatility dynamics and market-state information.

The own-stock memory coefficient is positive, whereas cross-sectional mean persistence and the own-stock persistence–VIX interaction enter negatively. The persistence–VIX coefficient is substantial in magnitude, ranging from -0.1453 to -0.1151 . Sectoral persistence coefficients are positive but small. These signs describe conditional associations in a model containing correlated predictors and interactions; they do not imply that higher persistence necessarily raises forecast volatility or prolongs stress.

Standardization aids comparison of coefficient magnitudes, but Lasso selection does not supply significance tests. Correlated inputs can also redistribute coefficients across related predictors. Consequently, retained aggregate features are not evidence that each block independently improves forecasting, and retention of own-stock memory does not contradict the weak performance of A_2 . The forecast comparisons, rather than coefficient magnitudes alone, establish the practical distinction between individual persistence and the joint state representation.

8.5. Regime Split

Table 10 reports MSE improvements relative to HAR by market regime and horizon. It distinguishes two questions: whether the combined forecasting system is useful during stress, and whether its persistence features add accuracy beyond HAR-X within those same states. The reported regime comparisons are descriptive and do not include regime-specific tests of C against HAR-X.

Table 10. Regime-Conditioned MSE Improvement (%) Relative to Model A

Panel A: HAR and persistence-augmented models								
Regime	<i>h</i>	A	A1	A2	A3	A4	A5	C
COVID (2020)	1	+0.00%	+11.84%	-1.51%	-0.36%	-1.41%	+9.13%	+6.87%
	5	+0.00%	+20.68%	+0.08%	-0.51%	-2.52%	+20.22%	+18.92%
	22	+0.00%	+13.56%	-8.27%	-2.56%	-7.08%	+8.68%	+7.24%
High VIX (Q4)	1	+0.00%	+11.44%	-1.30%	+0.32%	-0.51%	+11.21%	+11.83%
	5	+0.00%	+16.03%	-0.66%	+0.90%	-0.49%	+16.54%	+17.93%
	22	+0.00%	+5.58%	-3.63%	-0.83%	-2.18%	+2.67%	+1.07%
Low VIX (Q1)	1	+0.00%	+3.04%	-0.20%	+0.14%	+0.41%	+3.08%	+2.03%
	5	+0.00%	+5.60%	+0.88%	+1.75%	+1.24%	+5.76%	+6.59%
	22	+0.00%	+4.58%	-0.50%	+2.25%	+1.42%	+3.23%	+4.66%

Panel B: Shrinkage and tree-based models						
Regime	<i>h</i>	D_lasso	D_ridge	D_en	D_rf	D_gbm
COVID (2020)	1	+6.93%	+4.65%	+7.34%	-7.93%	-10.49%
	5	+16.48%	+13.77%	+16.19%	-0.42%	+5.47%
	22	+2.42%	+1.61%	+1.79%	-7.55%	-5.47%
High VIX (Q4)	1	+9.74%	+8.31%	+9.58%	+3.55%	-0.50%
	5	+15.03%	+13.52%	+14.91%	+6.32%	+5.58%
	22	+1.33%	+0.55%	+1.40%	-7.99%	-18.49%
Low VIX (Q1)	1	+1.53%	+1.16%	+1.53%	-2.84%	-21.87%
	5	+5.29%	+4.96%	+5.36%	-1.30%	-17.32%
	22	+3.76%	+3.95%	+3.79%	-7.27%	-20.35%

Notes: Out-of-sample MSE improvements on $\log RV^{PK}$ are measured relative to Model A within each regime. VIX quartiles are calculated over the out-of-sample evaluation window. The COVID window covers March–December 2020. Positive values indicate lower MSE than HAR within the regime. Panels A and B report different model groups using the same benchmark.

During high-VIX states, Model C improves on HAR by 11.83%, 17.93%, and 1.07% at the daily, weekly, and monthly horizons, compared with HAR-X gains of 11.44%, 16.03%, and 5.58%. The full model therefore adds 0.39 and 1.90 percentage points at the daily and weekly horizons and loses 4.51 percentage points at the monthly horizon on the HAR-relative improvement scale. The daily and weekly differences are of the size that Section 8.3 shows to be statistically indistinguishable from zero.

COVID is starker. Model C improves on HAR by 6.87%, 18.92%, and 7.24%, whereas HAR-X improves by 11.84%, 20.68%, and 13.56%. The full model therefore underperforms HAR-X at all three horizons during COVID. The weekly 18.92% gain is relative to HAR, not HAR-X, and is HAR-X's gain.

Gains over HAR also occur in calm states. At the weekly horizon, Model C improves by 6.59% in the low-VIX regime, versus 5.60% for HAR-X. The regime pattern therefore does not support the idea that persistence pays in stress: the combined model's large gains over HAR in stressed periods are the gains of the market-state controls.

Figure 6 compares the cross-sectional persistence state with VIX across leads and lags. Correlations are positive across the displayed range and are higher at negative than at positive lags. Under the plotted convention, positive lags indicate persistence leading VIX; the stronger negative-lag association is consistent with persistence responding to earlier market stress. The 750-day backward-looking estimation window smooths the persistence series and limits directional interpretation. This diagnostic does not establish persistence as a leading indicator or predict how long a stress episode will last.

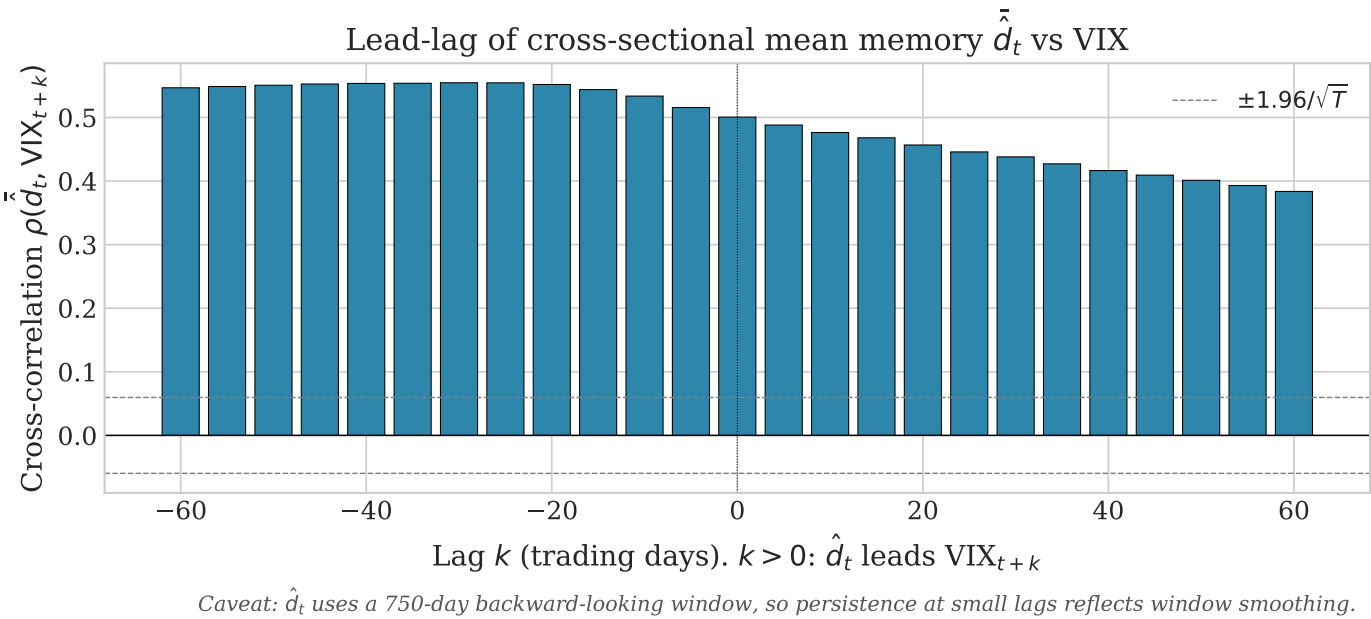


Figure 6. Lead–lag cross-correlation between cross-sectional mean memory $\hat{d}_t$ and VIX_{t+k} over ± 60 trading days. Positive k denotes persistence leading VIX. The persistence estimates use 750-day backward-looking windows.

Taken together, the regime comparisons do not support a state-dependent contribution of persistence beyond the market-state controls. They are consistent with persistence rising when stress rises, but they neither identify a propagation mechanism nor validate persistence as a direct measure or leading predictor of stress duration.

8.6. Sector Split

Table 11 reports weekly MSE improvements by GICS sector. Model C improves on HAR in all eleven sectors, with gains ranging from 4.19% in Information Technology to 12.72% in Materials. These comparisons establish breadth relative to HAR, but they do not attribute all sectoral gains to persistence.

HAR-X accounts for much of the improvement in every sector. Relative to HAR-X, Model C has lower MSE in eight sectors and higher MSE in Communication Services, Information Technology, and Utilities. The largest additional gains on the HAR-relative improvement scale occur in Health Care (2.13 percentage points), Materials (1.73), Industrials (1.41), and Consumer Staples (1.33). By contrast, the incremental changes in Financials and Real Estate are only 0.03 and 0.07 percentage points, despite their large total gains over HAR. Their performance should therefore not be presented as strong evidence that persistence adds information beyond market-state controls in funding-sensitive sectors.

Figure 7 extends the comparison with HAR across horizons. Every sector has its largest gain at the weekly horizon. At the monthly horizon the picture reverses: only Materials (+6.8%), Health Care (+3.6%), Industrials (+2.8%), and Real Estate (+2.0%) retain gains over HAR, while seven sectors, including Energy (−9.9%), Information Technology (−4.8%), and Communication Services (−4.6%), are worse than HAR. There is no long-horizon benefit across industries.

Sectoral heterogeneity is compatible with differences in common exposures and adjustment dynamics, but the table does not identify liquidity, leverage, or information-diffusion channels. Nor does heterogeneity itself rule out overfitting. The practical finding is narrower: the full feature set complements market-state controls in several sectors, while its incremental value is small or negative in others. Gradient Boosting underper-

Table 11. Sector-Level MSE Improvement (%) Relative to Model A at $h = 5$

Panel A: HAR and persistence-augmented models							
Sector	A	A1	A2	A3	A4	A5	C
Communication Services	+0.00%	+7.30%	-0.75%	+0.41%	+0.38%	+7.37%	+6.39%
Consumer Discretionary	+0.00%	+8.36%	-0.03%	+0.11%	+0.26%	+7.44%	+8.51%
Consumer Staples	+0.00%	+7.22%	+0.02%	+0.51%	+0.33%	+6.85%	+8.56%
Energy	+0.00%	+4.28%	+0.73%	+0.14%	+0.14%	+5.22%	+4.98%
Financials	+0.00%	+9.45%	+0.27%	+0.18%	+0.19%	+9.46%	+9.47%
Health Care	+0.00%	+6.62%	-0.60%	+0.61%	+0.11%	+7.12%	+8.75%
Industrials	+0.00%	+9.10%	-0.29%	+0.79%	+0.25%	+9.41%	+10.52%
Information Technology	+0.00%	+5.07%	-0.27%	+0.15%	+0.27%	+4.55%	+4.19%
Materials	+0.00%	+11.01%	-0.24%	+0.61%	+0.01%	+10.89%	+12.73%
Real Estate	+0.00%	+10.22%	+0.78%	-0.05%	-0.10%	+10.16%	+10.29%
Utilities	+0.00%	+8.07%	+1.45%	+0.23%	+0.22%	+6.95%	+7.62%

Panel B: Shrinkage and tree-based models			
Sector	D_lasso	D_rf	D_gbm
Communication Services	+5.77%	+0.54%	-14.59%
Consumer Discretionary	+6.96%	+1.66%	-9.57%
Consumer Staples	+6.48%	+0.88%	-10.83%
Energy	+1.80%	-2.77%	-13.06%
Financials	+7.82%	+4.28%	-6.72%
Health Care	+7.88%	+3.20%	-8.99%
Industrials	+8.31%	+3.40%	-8.14%
Information Technology	+3.49%	-0.63%	-14.57%
Materials	+11.28%	+4.10%	-12.16%
Real Estate	+10.51%	+5.67%	-7.98%
Utilities	+9.15%	+2.58%	-4.61%

Notes: Out-of-sample MSE improvements on $\log RV^{PK}$ at $h = 5$ are measured relative to Model A and pooled across stocks within each GICS sector over the evaluation window. Panels A and B report different model groups using the same benchmark.

forms HAR in every sector at the weekly horizon, whereas the relative performance of shrinkage methods varies by sector.

8.7. Robustness

The final step in the empirical analysis is to determine whether the forecasting improvements remain stable across alternative estimation procedures, rolling-window choices, volatility proxies, liquidity conditions, and benchmark specifications.

Table 12 summarizes a broad robustness battery for the headline comparison between the full persistence-augmented model and the HAR benchmark at the weekly forecasting horizon. The reported variants retain positive gains against HAR, subject to the benchmark and measurement qualifications discussed below. Replacing the GPH persistence estimator throughout the feature set with the local-Whittle estimator produces forecasting improvements almost identical to the headline specification: +8.21% versus the baseline +8.25%. This result is important because it demonstrates that the forecasting gains are not driven by the particular semiparametric estimator used to construct the persistence variables.

The rolling-window robustness checks produce similarly stable results. Using a shorter 500-day rolling estimation window yields approximately +9.13% forecast improvement, while using a longer 1000-day window yields approximately +8.68%. Both forecasting gains remain statistically significant. These findings indicate that the forecasting gains are not artifacts of a particular rolling-window length. Persistence-based forecasting variables remain informative across substantially different temporal smoothing structures.

The volatility-proxy robustness checks are also revealing. Replacing the Parkinson realized-variance target with the logarithm of future mean squared returns, a substantially

Table 12. Robustness Summary: Model C Relative to Model A at $h = 5$

Variant	MSE	% Δ vs A	HLN DM- t
Headline (Model C, full panel)	0.3335	+8.25%	+3.88
Inference: plain DM (illustrative)	0.3335	+8.25%	+3.88 (plain +23.99)
Benchmark: GARCH(1,1) on returns	0.7119	-95.87%	-20.64
Liquidity: low-liquidity half	0.3342	+8.95%	+4.26
Liquidity: high-liquidity half	0.3327	+7.52%	+3.33
Estimator: $\hat{d}_{LW}$ replaces $\hat{d}_{GPH}$	0.3336	+8.21%	+3.85
Rolling window: 500 days	0.3302	+9.13%	+4.00
Rolling window: 1000 days	0.3319	+8.68%	+3.63
Target: log mean future squared returns	0.9093	+4.19%	+3.15
Regime: high VIX (Q_4)	–	+17.94%	–
Regime: low VIX (Q_1)	–	+6.57%	–
Regime: COVID 2020	–	+18.88%	–

Notes: Unless indicated otherwise, the checks compare Model C with Model A at $h = 5$ using the log RV^{PK} target. Liquidity groups are formed by splitting at the cross-sectional median of full-sample mean inverse dollar volume. The estimator variant replaces GPH with local-Whittle estimates throughout the persistence block, including dynamics and interactions. Window variants reestimate rolling persistence before reconstructing the features. The target variant uses the log of future mean squared returns and refits both A and C. The GARCH(1,1) row reports a constant-mean model fitted to log returns and refitted every 20 sample steps, with the log of mean forecast conditional variance evaluated against the Parkinson target. Its return-variance scale differs from the range-based target, complicating interpretation of its relative loss. Under the proxy-robust QLIKE loss of Patton (2011), the GARCH(1,1) HLN-corrected DM statistic against Model A narrows to -2.80 ($p = 0.005$), versus -20.30 under MSE in the column above. Regime rows reproduce the $h = 5$ comparisons in Table 10. The plain DM statistic treats pooled cells as independent; the HLN statistic uses panel-aware aggregation and the Harvey–Leybourne–Newbold correction.

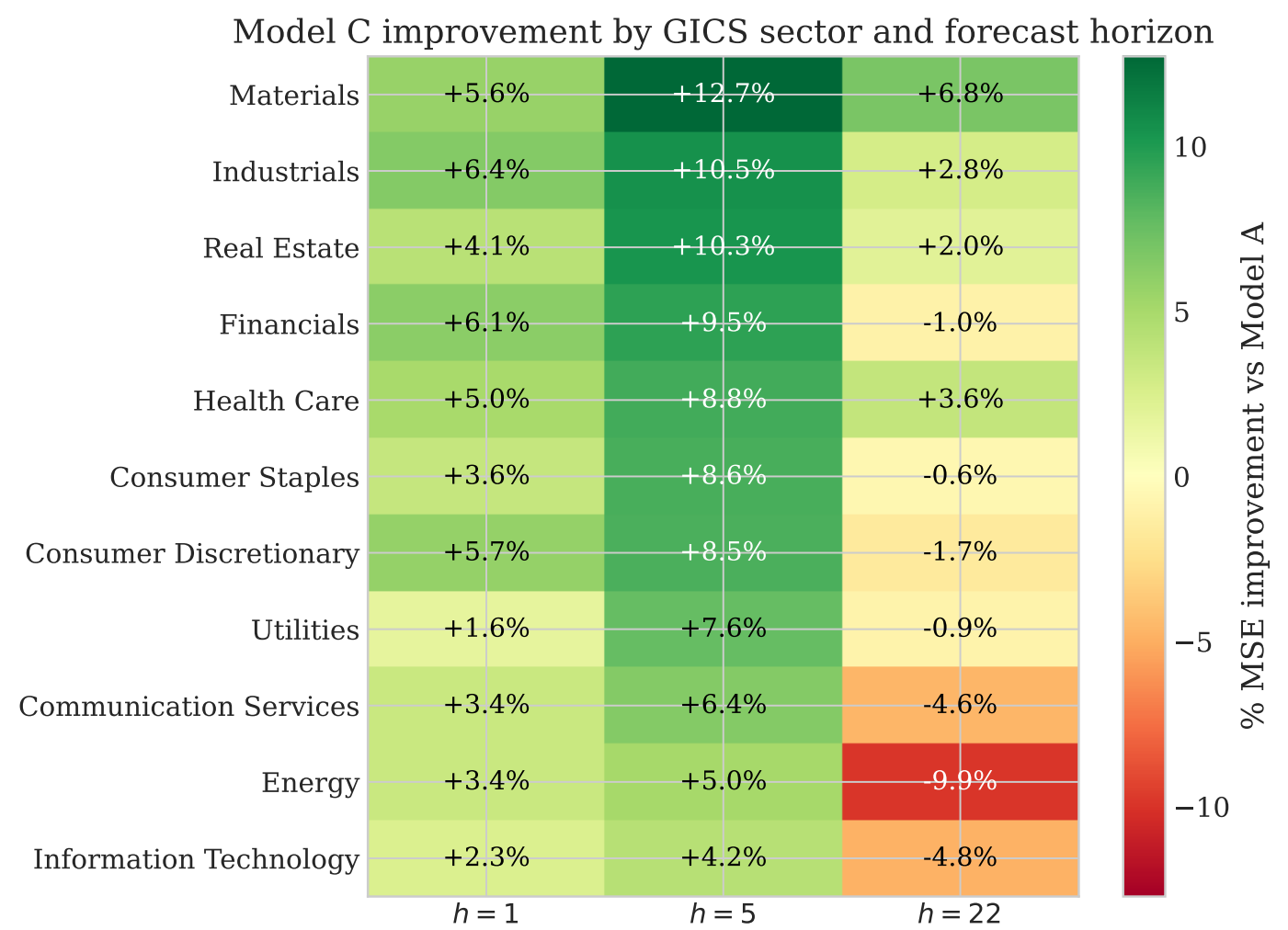


Figure 7. Percentage MSE improvement of Model C over Model A by GICS sector and forecast horizon. Sectors are sorted by the $h = 5$ gain.

noisier volatility proxy, reduces the magnitude of the forecasting gains but still produces statistically significant improvement: +4.19%. This result suggests that the persistence-based forecasting gains are not specific to the Parkinson volatility estimator itself. The smaller gain is consistent with sensitivity to the volatility proxy, but the comparison does not isolate measurement noise as its cause.

Liquidity splits show gains over HAR in both groups: 8.95% in the low-liquidity half and 7.52% in the high-liquidity half. The larger gain in the former is consistent with a liquidity-related interpretation, but the split does not identify a liquidity mechanism. These groups use a static full-sample liquidity classification and are retrospective diagnostics, not real-time trading rules.

The GARCH(1,1) benchmark row warrants comment. A conventional GARCH(1,1) model estimated directly on returns performs substantially worse than the HAR benchmark when evaluated against the Parkinson range-based variance target: −95.9% relative performance under mean squared error loss. Under QLIKE, the reported HLN DM statistic is −2.96. This statistic is not directly comparable to a percentage MSE change, since the loss functions use different scales. Comparisons between returns-based conditional variance models and range-based proxies involve substantial scale mismatch and measurement differences. The paper therefore emphasizes that the forecasting comparison should

primarily be interpreted within the realized-volatility forecasting environment rather than as a definitive rejection of all conditional-variance models.

Finally, the robustness analysis again illustrates the importance of correct statistical inference. The naive pooled-cell Diebold–Mariano statistic ignoring cross-sectional dependence equals approximately +23.99, whereas the properly panel-aware HLN-corrected statistic equals approximately +3.88. This difference illustrates how apparent statistical significance may be inflated when within-date cross-sectional dependence is ignored.

Taken together, the robustness checks show that the combined model's weekly gains against HAR persist under the reported estimator, window, proxy, and liquidity variations. They compare C with HAR; the comparison that matters, C against HAR-X, is in Section 8.3 and is not favorable. The checks also do not resolve sample selection or structural-break identification.

9. Economic Interpretation and Implications

The forecasting evidence does not support using persistence states as additional predictors once implied-volatility indices are available; it does support treating the cross-sectional persistence state as a descriptive indicator of market conditions. The portfolio exercise applies the existing forecasts and tests whether the small forecast differences translate into risk-adjusted performance; the broader implications for stress monitoring remain potential uses rather than separately validated results.

9.1. Volatility-Managed Portfolios

Following [Moreira and Muir \(2017\)](#), the paper constructs equity portfolios that scale exposure inversely with forecast variance. For stock i at date t , the managed five-day return is

$$f_{m,i,t} = \frac{c_{m,i}}{\hat{\sigma}_{m,i,t}^2} r_{i,t \rightarrow t+5}, \quad (17)$$

where $\hat{\sigma}_{m,i,t}^2 = \exp(\hat{y}_{m,i,t})$ is the exponentiated log-variance forecast and $c_{m,i,t}$ is a stock- and model-specific normalization constant that equates the variance of the managed and unmanaged stock returns over the expanding window of weeks strictly before t , after a 52-week warm-up, so that the strategy is implementable in real time; the [Moreira and Muir \(2017\)](#) constant computed over the full evaluation sample is reported as a robustness row. Portfolio returns use raw (unwinsorized) log returns. The Parkinson proxy is the forecast target; this construction should not be interpreted as directly forecasting the conditional variance of portfolio returns.

Table 14 shows similar Sharpe ratios over the 593-week evaluation sample that follows the normalization warm-up: 0.61 for the unmanaged portfolio, 0.63 for HAR-X management, and 0.66 for Model C management. Model C has a smaller maximum drawdown than both comparators (−17.4% versus −18.2% and −31.9%) and a marginally higher certainty-equivalent return (4.19% versus 3.93% and 3.34%). The full-sample-constant construction of [Moreira and Muir \(2017\)](#) gives a lower Sharpe ratio (0.59) for the same forecasts, so the real-time version does not flatter the strategy.

During COVID, Model C management produces a Sharpe ratio of 1.41, compared with 1.11 for HAR-X management and 0.50 for the unmanaged portfolio. Its certainty-equivalent return is 12.04%, versus 8.57% and −11.12%, respectively. Its COVID maximum drawdown is larger than that of HAR-X management in absolute magnitude (4.73% versus 4.04%). In high-VIX states, the Sharpe ratios for Model C and HAR-X management are 0.78 and 0.71; in low-VIX states, they are 1.15 and 1.14.

Table 13 asks whether any of these differences is distinguishable from zero. None is. The annualized Sharpe difference between Model C and HAR-X management is +0.03

Table 13. Volatility-Managed Portfolios: Tests of Sharpe-Ratio Differences

Regime	Comparison	Weeks	Sharpe (1)	Sharpe (2)	Δ Sharpe	HAC t	Bootstrap p
Full sample	Model C vs HAR-X	593	0.66	0.63	+0.03	+1.02	0.292
Full sample	Model C vs Unmanaged	593	0.66	0.61	+0.05	+0.29	0.786
Full sample	HAR-X vs Unmanaged	593	0.63	0.61	+0.03	+0.16	0.877
Low VIX (Q1)	Model C vs HAR-X	150	1.16	1.14	+0.02	+0.46	0.620
Low VIX (Q1)	Model C vs Unmanaged	150	1.16	1.06	+0.10	+0.80	0.533
Low VIX (Q1)	HAR-X vs Unmanaged	150	1.14	1.06	+0.08	+0.63	0.625
High VIX (Q4)	Model C vs HAR-X	149	0.78	0.71	+0.07	+1.06	0.291
High VIX (Q4)	Model C vs Unmanaged	149	0.78	0.74	+0.04	+0.12	0.939
High VIX (Q4)	HAR-X vs Unmanaged	149	0.71	0.74	-0.03	-0.08	0.948
COVID 2020	Model C vs HAR-X	43	1.43	1.13	+0.30	+1.43	0.369
COVID 2020	Model C vs Unmanaged	43	1.43	0.51	+0.92	+1.11	0.411
COVID 2020	HAR-X vs Unmanaged	43	1.13	0.51	+0.62	+0.88	0.485

Notes: Weekly (five-trading-day, non-overlapping) portfolio returns on the forecast sample stride; Sharpe ratios annualised by $\sqrt{252/5}$. HAC t is the Ledoit–Wolf (2008) delta-method statistic for the difference in Sharpe ratios; bootstrap p is from a studentised circular block bootstrap (block length 5, 2000 resamples). Significance of the HAC statistic: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Regime rows use the same VIX quartiles and COVID window as the regime table.

over the full sample (Ledoit–Wolf HAC $p = 0.31$, bootstrap $p = 0.29$), +0.07 in high-VIX weeks ($p = 0.29$), and +0.30 in COVID ($p = 0.15$, bootstrap $p = 0.37$, on 43 weekly observations). The COVID gap is the largest number in the table and the least informative: 43 weeks cannot separate Sharpe ratios of 1.41 and 1.11.

These portfolio comparisons need not reproduce the MSE ranking. During COVID at $h = 5$, HAR-X has lower forecast MSE than Model C, yet the Model C-managed portfolio has the higher Sharpe ratio. Forecast loss and portfolio performance evaluate different objects: inverse-variance weights and realized returns determine the latter. The portfolio result is an application-level comparison, not proof that persistence improves COVID weekly forecast accuracy or uniquely causes the Sharpe difference.

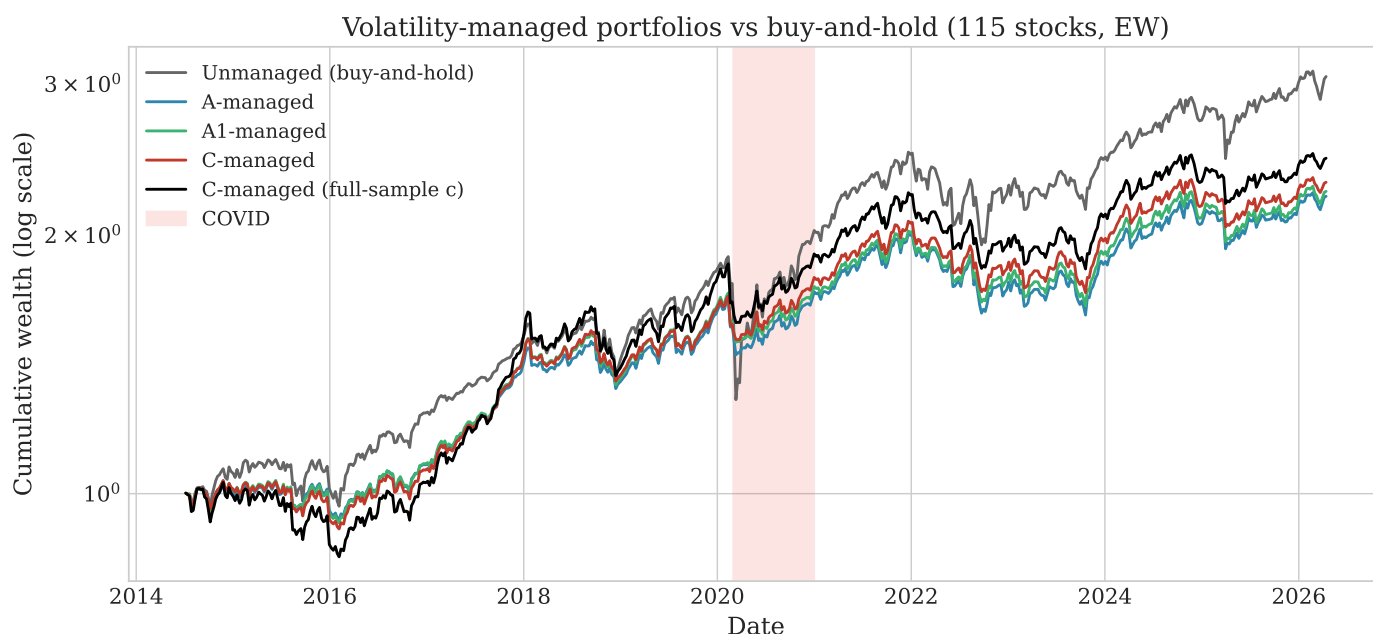


Figure 8. Cumulative wealth (log scale) of the unmanaged equal-weight portfolio, the three volatility-managed variants with real-time normalization, and Model C management with the full-sample [Moreira and Muir \(2017\)](#) constant, over the evaluation window that follows the 52-week normalization warm-up. The shaded band marks March–December 2020.

Table 14. Volatility-Managed Portfolios: Risk-Adjusted Performance (Moreira-Muir 2017 Construction)

Regime	Portfolio	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	CER
Full sample	Unmanaged (buy-and-hold)	+9.51%	+15.71%	+0.61	-31.86%	+3.34%
	A-managed	+6.78%	+11.00%	+0.62	-19.57%	+3.75%
	A1-managed	+6.89%	+10.89%	+0.63	-18.21%	+3.93%
	C-managed	+7.10%	+10.79%	+0.66	-17.44%	+4.19%
	C-managed (full-sample c)	+7.65%	+12.88%	+0.59	-18.51%	+3.50%
Low VIX (Q1)	Unmanaged (buy-and-hold)	+8.00%	+7.59%	+1.05	-15.38%	+6.56%
	A-managed	+9.19%	+8.83%	+1.04	-17.68%	+7.24%
	A1-managed	+10.52%	+9.26%	+1.14	-17.44%	+8.38%
	C-managed	+10.93%	+9.49%	+1.15	-17.94%	+8.68%
	C-managed (full-sample c)	+11.92%	+12.53%	+0.95	-25.09%	+8.00%
High VIX (Q4)	Unmanaged (buy-and-hold)	+18.55%	+25.13%	+0.74	-29.18%	+2.76%
	A-managed	+9.31%	+13.06%	+0.71	-15.67%	+5.05%
	A1-managed	+8.40%	+11.84%	+0.71	-13.65%	+4.90%
	C-managed	+9.03%	+11.57%	+0.78	-13.29%	+5.68%
	C-managed (full-sample c)	+9.97%	+12.64%	+0.79	-13.99%	+5.98%
COVID 2020	Unmanaged (buy-and-hold)	+16.65%	+33.33%	+0.50	-15.25%	-11.12%
	A-managed	+10.70%	+11.29%	+0.95	-3.98%	+7.51%
	A1-managed	+11.03%	+9.90%	+1.11	-4.04%	+8.57%
	C-managed	+14.78%	+10.47%	+1.41	-4.73%	+12.04%
	C-managed (full-sample c)	+15.09%	+11.88%	+1.27	-5.20%	+11.57%

Notes: Equal-weight portfolio of 115 stocks, weekly ($h = 5$ trading days) rebalancing aligned to the forecast sample stride. The volatility-managed portfolio uses position weight $w_{m,i,t} = c_{m,i} / \hat{\sigma}_{m,i,t}^2$ where $\hat{\sigma}_{m,i,t}^2 = \exp(\hat{y}_{m,i,t})$ is the model- m forecast of mean RV^{PK} over $[t+1, t+5]$. The constant $c_{m,i,t}$ equates the variance of the managed and unmanaged stock returns over the expanding window of weeks before t (52-week warm-up), so the strategy is implementable in real time; the 'full-sample c ' row uses the Moreira–Muir constant computed over the whole evaluation sample. Portfolio returns use raw (unwinsorised) log returns. Annualized assuming 252/5 five-day periods per year. CER computed for a mean-variance investor with risk aversion $\gamma = 5$.

Figure 8 displays the corresponding wealth paths over the evaluation period following the initial training window. It does not provide out-of-sample evidence for the global financial crisis, which predates the stated forecast evaluation start. The exercise illustrates the consequences of forecast-based exposure scaling rather than identifying how uncertainty is transmitted or resolved.

The normalization is real-time and Table 13 supplies the inference, but transaction costs, financing, and exposure constraints are not modeled. Accordingly, the exercise is evidence about the reported strategy construction, not a verified net-of-cost investment recommendation.

9.2. Implications for Risk Monitoring

The cross-sectional persistence state rises during the global financial crisis and COVID and has a contemporaneous correlation of approximately 0.50 with VIX. These full-sample diagnostics motivate monitoring dependence alongside volatility levels. They do not establish that persistence predicts the onset of stress or the time until conditions normalize. The lead–lag correlations and the smoothing induced by rolling estimation do not establish a directional forecasting relationship.

For risk management, the immediate implication is that a persistence state is a useful monitoring variable but not a forecasting input once implied-volatility indices are available. It may still motivate scenarios involving more persistent volatility or sectorally concentrated dependence. Applications to crisis-duration forecasting, systemic early warning, multivariate risk, and volatility trading remain extensions requiring their own evaluation. The current results do not validate those uses.

9.3. Implications for Forecasting Design

The comparison across estimators favors a structured linear use of the feature set over the evaluated tree-based alternatives. The relevant methodological lesson is conditional: interpretable feature construction can be useful without a highly flexible estimator in this panel and forecasting protocol. It is not a general claim that nonlinear methods cannot forecast volatility, nor does the comparison alone establish whether sample size, tuning, or estimation instability explains their performance. HAR-X remains a strong benchmark throughout.

10. Conclusion

This paper evaluates volatility persistence as a financial state variable rather than solely as a stock-specific fractional parameter. Across 115 equities drawn from the S&P 500 over November 2001–April 2026, the volatility proxies display persistent dependence and low estimated Hurst exponents. The cross-sectional mean GPH and local-Whittle estimates are 0.451 and 0.440, respectively, and the mean Hurst estimate is approximately 0.063. These diagnostics are consistent with persistence at longer scales and local irregularity, without identifying a unique latent volatility process or establishing stationarity in every rolling window.

The main finding is that the forecasting content of this state is already contained in the market-state controls. The own-stock persistence and roughness block adds nothing to HAR. The shared persistence state, combined with VIX and MOVE, improves on HAR by 4.5% and 8.3% at the daily and weekly horizons, but HAR-X alone achieves 5.1% and 7.7%, and at the monthly horizon neither the full model nor any persistence block improves on HAR once the training window is restricted to information available at the forecast origin.

Tested directly, the full model is never significantly more accurate than HAR-X (Diebold–Mariano statistics of -0.74 , $+0.48$, and -2.44). Clark–West tests show that the

persistence regressors do carry population information, but three specifications fixed in advance that add persistence to HAR-X directly, including one in which persistence modulates the HAR weights, all fail to lower forecast error. Pooled estimation across stocks recovers a gain of two tenths of one percent at the daily horizon, the persistence state does not forecast the slope of the future variance term structure, and it adds nothing to VIX and MOVE at the market level. The duration interpretation that motivates persistence as a state variable is therefore not supported as a forecastable quantity.

The same holds across regimes and sectors. The full model's large gains over HAR in stressed periods are HAR-X's gains; it trails HAR-X at every horizon during COVID; and its weekly sectoral advantages are not accompanied by significant differences from HAR-X.

The volatility-managed portfolio exercise, with real-time normalization, gives Model C management a COVID Sharpe ratio of 1.41 versus 1.11 for HAR-X management and 0.66 versus 0.63 over the full sample; neither difference is statistically significant ($p = 0.15$ and 0.31). The estimator comparisons favor the structured linear specification over the evaluated tree-based procedures within this setting, not linear methods in general.

Several limitations define the scope of the findings. The panel is the S&P 500 membership of April 2026 and is subject to survivorship and selection effects; range-based volatility measurement is not equivalent to observing latent volatility; and rolling estimates can reflect regime changes as well as fractional dependence, with some values outside the stationary range. The persistence state is a coherent descriptive indicator of market stress, but no causal mechanism is identified, and the VIX lead-lag diagnostic documents an association, not a validated leading indicator.

Future work could assess point-in-time constituent panels, richer high-frequency measurements, other asset classes, and settings without a liquid implied-volatility index, where the information that persistence shares with the VIX would have to come from persistence itself. The present contribution is the characterization: a shared persistence state describes market conditions well, and its forecasting content for stock-level variance is already priced in implied volatility.

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References

- Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and Forecasting Realized Volatility. *Econometrica*, 71(2), 579–625.

- Baillie, R. T., Bollerslev, T., & Mikkelsen, H. O. (1996). Fractionally Integrated Generalized Autoregressive Conditional Heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30.
- Barndorff-Nielsen, O. E., & Shephard, N. (2002). Econometric Analysis of Realized Volatility and Its Use in Estimating Stochastic Volatility Models. *Journal of the Royal Statistical Society: Series B*, 64(2), 253–280.
- Bennedsen, M., Lunde, A., & Pakkanen, M. S. (2022). Decoupling the Short- and Long-Term Behavior of Stochastic Volatility. *Journal of Financial Econometrics*, 20(5), 961–1006.
- Bollerslev, T. (1986). Generalized Autoregressive Conditional Heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.
- Clark, T. E., & West, K. D. (2007). Approximately Normal Tests for Equal Predictive Accuracy in Nested Models. *Journal of Econometrics*, 138(1), 291–311.
- Corsi, F. (2009). A Simple Approximate Long-Memory Model of Realized Volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- Diebold, F. X., & Mariano, R. S. (1995). Comparing Predictive Accuracy. *Journal of Business and Economic Statistics*, 13(3), 253–263.
- Ding, Z., Granger, C. W. J., & Engle, R. F. (1993). A Long Memory Property of Stock Market Returns and a New Model. *Journal of Empirical Finance*, 1(1), 83–106.
- Engle, R. F. (1982). Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. *Econometrica*, 50(4), 987–1007.
- Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, 29(5), 1189–1232.
- Gatheral, J., Jaisson, T., & Rosenbaum, M. (2018). Volatility Is Rough. *Quantitative Finance*, 18(6), 933–949.
- Geweke, J., & Porter-Hudak, S. (1983). The Estimation and Application of Long Memory Time Series Models. *Journal of Time Series Analysis*, 4(4), 221–238.
- Granger, C. W. J., & Joyeux, R. (1980). An Introduction to Long-Memory Time Series Models and Fractional Differencing. *Journal of Time Series Analysis*, 1(1), 15–29.
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical Asset Pricing via Machine Learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the Equality of Prediction Mean Squared Errors. *International Journal of Forecasting*, 13(2), 281–291.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer.
- Hoerl, A. E., & Kennard, R. W. (1970). Ridge Regression: Biased Estimation for Nonorthogonal Problems. *Technometrics*, 12(1), 55–67.
- Holm, S. (1979). A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Hosking, J. R. M. (1981). Fractional Differencing. *Biometrika*, 68(1), 165–176.
- Jarque, C. M., & Bera, A. K. (1980). Efficient Tests for Normality, Homoscedasticity and Serial Independence of Regression Residuals. *Economics Letters*, 6(3), 255–259.
- Ledoit, O., & Wolf, M. (2008). Robust Performance Hypothesis Testing with the Sharpe Ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Mikosch, T., & Stărică, C. (2004). Nonstationarities in Financial Time Series, the Long-Range Dependence, and the IGARCH Effects. *Review of Economics and Statistics*, 86(1), 378–390.
- Moreira, A., & Muir, T. (2017). Volatility-Managed Portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Müller, U. A., Dacorogna, M. M., Davé, R. D., Olsen, R. B., Pictet, O. V., & von Weizsäcker, J. E. (1997). Volatilities of Different Time Resolutions—Analyzing the Dynamics of Market Components. *Journal of Empirical Finance*, 4(2–3), 213–239.
- Newey, W. K., & West, K. D. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica*, 55(3), 703–708.
- Parkinson, M. (1980). The Extreme Value Method for Estimating the Variance of the Rate of Return. *Journal of Business*, 53(1), 61–65.
- Patton, A. J. (2011). Volatility Forecast Comparison Using Imperfect Volatility Proxies. *Journal of Econometrics*, 160(1), 246–256. <https://doi.org/10.1016/j.jeconom.2010.03.034>.

- Robinson, P. M. (1995). Gaussian Semiparametric Estimation of Long Range Dependence. *Annals of* 1314
Statistics, 23(5), 1630–1661. 1315
- Tibshirani, R. (1996). Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical* 1316
Society: Series B, 58(1), 267–288. 1317
- Zou, H., & Hastie, T. (2005). Regularization and Variable Selection via the Elastic Net. *Journal of the* 1318
Royal Statistical Society: Series B, 67(2), 301–320. 1319